RareAgentBench: A Multi-Pillar, Contamination-Controlled Benchmark of LLM Agents for Rare-Disease Diagnosis

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Abstract

1 Abstract

Rare disease diagnostic AI agents have proliferated (8+ systems in 2024-2026), yet no shared benchmark exists; each agent paper evaluates on an ad-hoc subset, making cross-system claims unverifiable. We introduce RareAgentBench, an agent-native benchmark spanning five capability pillars (phenotype extraction, phenotype-only DDx, genotype-aware DDx, family-aware DDx, clinical-communication faithfulness) on a layered dataset (Phenopacket-Store n=10,051; RareBench HF 1,122; RareArena RDS 72,661; MIMIC-IV rare-disease slice n=956; post-cutoff PMC OA holdout n=200). We evaluate 8 agent systems (DeepRare, MDAgents, MedAgents, AgentClinic, MAI-DxO, RDMA, VC-RDAgent, LIRICAL) against 3 LLM no-scaffolding controls and one classical baseline, with all hypotheses (H1–H11) and ablations (A1–A12) pre-registered.

Key findings: (1) classical/offline baselines (LIRICAL 0.46, VC-RDAgent 0.44 R@1) exceed every scaffolded LLM agent on HPO-input datasets (best LLM cell 0.33 on Phenopacket-Store, a 13 pp gap); (2) multi-agent scaffolding gives only a small, dataset-dependent gain (≈2–5 pp R@1 over single-LLM controls), not the uniform boost prior work implies; (3) DeepSeek V4-Flash is ~10× cheaper than Gemini Flash but trades off accuracy (−2 to −9 pp on structured input, −11 to −16 pp on free-text) — cost-efficient, not quality-equivalent; (4) GPT-5 with `reasoning_effort=minimal` carries the highest cost (~34× V4-Flash) without a consistent accuracy edge — competitive on some scaffolds yet collapsing on dialogue (−14 pp on AgentClinic), illustrating frontier-reasoning models' brittleness under reasoning-disabled regimes; (5) an ORPHA-variant evaluation channel adds ~20 pp R@1 universally — *not* DeepRare-specific. We release the harness, canonical case schema, per-agent adapter shims, full per-cell receipts, and a static-site leaderboard.

2 Introduction

2.1 The phenomenon — agent proliferation in rare disease

Rare disease diagnosis is undergoing a diagnostic-agent renaissance. In 2024–2026 alone, eight distinct agent systems were proposed that attack the problem with markedly different architectures: medical multi-agent debate (MDAgents [1], MedAgents [2], MAI-DxO [3]), OSCE dialogue simulation (AgentClinic [4]), domain-specialised retrieval + reasoning (DeepRare [5]), HPO-conditional fusion (VC-RDAgent [6]), phrase-mining (RDMA [7]), and classical Bayesian likelihood (LIRICAL [8]). Each system reports R@1 in the 0.3–0.7 range on its own evaluation set.

2.2 The gap — no shared benchmark

But no shared benchmark exists. Each paper evaluates on an ad-hoc subset: MDAgents on MedQA-Rare, MAI-DxO on NEJM clinicopathologic cases, DeepRare on a 9-source proprietary mix, RareBench on a 1.1k HPO subset, etc. Cross-system comparison is impossible. A recent systematic review (Garrett et al., *npj Digital Medicine* 2026) audited 19 LLM rare-disease studies and flagged (i) absence of pre-registration in all but one, (ii) high contamination risk across all 19, and (iii) R^2 = 0.55 between disease prevalence and reported R@1 — a strong signal that selection bias dominates published numbers. Without a shared evaluation infrastructure, "agent X beats agent Y" is unverifiable.

2.3 Why agent-native, not LLM++

One could argue an LLM-only benchmark suffices. We disagree on three grounds. First, the diagnostic agent designs we audit differ on *non-LLM* axes — RAG pipelines (DeepRare), panel orchestration (MAI-DxO), OSCE dialogue (AgentClinic), classical fusion (VC-RDAgent) — that LLM-only benchmarks cannot disentangle from backbone choice. Second, the *evaluation surface* matters: pure accuracy ignores faithfulness, cost, and pass-k reliability, all of which matter in clinical deployment. Third, classical / offline baselines (LIRICAL Bayesian, VC-RDAgent Stage 1) are competitive with LLM agents on HPO-input datasets — a result that *no* prior benchmark would have surfaced because none of them run classical baselines side-by-side with the LLM lineup.

2.4 Our approach

We introduce RareAgentBench, a benchmark with three structural commitments:

Five capability pillars: phenotype extraction (P1), phenotype-only DDx (P2), genotype-aware DDx (P3), family-aware DDx (P4, deferred to v2), and clinical-communication faithfulness (P5). Each pillar surfaces different agent capabilities; collapsing all into a single accuracy number erases >30% of variance, as our Section B.3 Spearman $\rho \approx 0.36$ between R@1 and faithfulness demonstrates.

Layered dataset: Phenopacket-Store (10,051 cases) → RareBench HF (1,122) → RareArena RDS (72,661) → MIMIC-IV rare-disease slice (956) → PMC OA post-cutoff holdout (200). The layering spans difficulty, contamination risk, input modality (HPO vs free-text), and prevalence distribution. Holdout case excerpts are post-LLM-training-cutoff (after 2026-02), eliminating the contamination concern that plagues prior work.

Pre-registered hypotheses + ablations: H1–H11 + A1–A12 frozen at OSF prior to holdout unblinding. To our knowledge this is the first pre-registered rare-disease diagnostic-AI evaluation.

2.5 Key findings preview

We evaluate 8 agent systems against 3 LLM controls (Gemini 3 Flash, DeepSeek V4-Flash, GPT-5 *reasoning_effort=minimal*) and 1 classical baseline across the four pre-holdout layers (Phase 4a, N=100/dataset, $54 cumulative cost — code, data, and per-cell receipts open-sourced).

Five findings:

1. Classical/offline baselines exceed every scaffolded LLM agent on HPO-input datasets: LIRICAL (Bayesian) R@1 = 0.46 [0.42–0.51]; VC-RDAgent (offline IC+Poincaré) 0.44 [0.40–0.48]; best LLM cell 0.33 (MedAgents × Gemini, N=500) on Phenopacket-Store — a 13 pp gap.
2. Multi-agent scaffolding gives only a small, dataset-dependent gain (≈2–5 pp R@1) over single-LLM controls, not the uniform boost prior work implies (medagents 0.33 vs llm_control 0.31, Phenopacket-Store / Gemini, N=500; within overlapping CIs).
3. DeepSeek V4-Flash is ~10× cheaper than Gemini Flash but trades off accuracy ($0.11/$0.22 vs higher Gemini pricing; R@1 −2 to −9 pp on structured input, −11 to −16 pp on free-text) — cost-efficient, not quality-equivalent.
4. GPT-5 with `reasoning_effort=minimal` is the most expensive backbone (~34× V4-Flash) with no consistent accuracy edge — best on MedAgents yet collapsing −14 pp on AgentClinic dialogue — exposing frontier reasoning models' brittleness under reasoning-disabled regimes.
5. Variant channel adds ~20 pp R@1 to *any* agent that ingests structured variants (Phase 3.2 P3 pilot: llm_control 0.26 → 0.46, deeprare 0.22 → 0.38) — *not* DeepRare-specific.

2.6 Contributions

- **RareAgentBench benchmark**: 5 pillars × 4 layers × 11 systems × 3 backbones, pre-registered + open-sourced.
- **Per-agent adapter shims** (3,485 LOC): unified `CanonicalCase` input contract + subprocess isolation, with per-baseline reproducibility documentation.
- **Reproducibility receipts**: per-cell run-id + OpenRouter request-id
    - dollar cost; full Phase 4a matrix (7,581 predictions) released.
- **Pre-registered statistical protocol**: Holm–Bonferroni for H1–H11, bootstrap 95% CIs, LLM-judge self-preference detection (Gemini-judge → Claude-judge agreement floor).
- **Static-site leaderboard** for community ratchet.

3 Related Work

3.0.1 Rare Disease LLM Benchmarks

Existing benchmarks are LLM-centric, not agent-native. Among nine specialized rare-disease diagnostic benchmarks published 2023-2026, eight evaluate base LLMs under prompting / few-shot / RAG, and one is hybrid . RareBench [Chen et al., KDD 2024] establishes the modern protocol with 2,764 patients across five subsets (RAMEDIS, MME, HMS, LIRICAL, PUMCH-ADM) and reports Recall@1/3/10 plus median rank. RareArena [Zhao et al., Lancet Digital Health 2025] scales to 49,760 free-text case reports across 4,597 Orphanet disorders. Phenopacket-Store [Danis et al., HGG Adv 2025] curates 7,552 GA4GH Phenopackets covering 481 OMIM diseases. Reese et al. [Eur J Hum Genet 2026] and Chimirri et al. [eBioMedicine 2025] add 5,213 and 4,917 multi-language Phenopackets respectively. MIMIC-RD [Wu et al., arXiv 2026] introduces real EHR free-text via 145 admissions mined from MIMIC-IV. All evaluate static input→output LLM prompting; none expose interactive tool APIs, cost/latency dimensions, reasoning-trace evaluation, or pass^k reliability — the dimensions that characterize agent systems. A 2026 systematic review independently flags this gap, observing that all 19 LLM rare-disease evaluations it surveys carry high data-contamination risk, no prevalence stratification, and no agent-process metrics.

3.0.2 Agent Benchmarks in Other Domains

Agent benchmarks elsewhere have matured well past static evaluation; rare-disease benchmarks have not. τ-bench [Yao et al., 2024] formalized `pass^k` reliability for tool-using agents, finding GPT-4o below 25% under k=8 i.i.d. retries on retail scenarios. AgentBoard [Ma et al., NeurIPS 2024] introduces Progress Rate as a partial-credit metric with Pearson r > 0.95 against human judgment across nine task domains. SWE-bench [Jimenez et al., ICLR 2024] sets the precedent for issue-resolution as the headline metric in agent benchmarks. In medicine, MedAgentBench [Jiang et al., NEJM AI 2025] introduces 100 tool-augmented FHIR query tasks but does not cover rare-disease diagnosis. MedHELM [Patel et al., 2025] extends HELM's scenario $\times$ metric matrix to 35 medical scenarios, with bias/fairness as cross-cutting evaluation lenses rather than separate pillars. We adopt three design patterns from these: (i) bias as a cross-cutting lens rather than a pillar (per HELM/MedHELM), (ii) `pass^k` reliability and Cost-Normalized Accuracy from the τ-bench / CLEAR lineage, and (iii) Progress-Rate-style partial credit for multi-stage agents.

3.0.3 Rare-Disease and Medical Agent Systems

Eight agent systems exist for rare disease or transferable to it; none share a benchmark. DeepRare [Yao et al., Nature 2026] is the current SOTA, integrating 40+ tools (HPO, Orphanet, OMIM, PubMed, web search, variant analyzers) under a central-host architecture with reflection; the authors evaluate on nine ad-hoc datasets totaling 6,401 patients. MAI-DxO [Nori et al., arXiv 2506.22405] (Microsoft Diagnostic Orchestrator) coordinates an eight-role panel with sequential test-ordering and a `budgeted` mode that caps per-case spend. RareAgents [Chen et al., AAAI 2026] applies multi-disciplinary team (MDT) reasoning with a specialty memory; RDMA [Wu et al., arXiv 2507.15867] specializes in EHR mining and HPO extraction; VC-RDAgent uses offline Poincaré-embedded HPO knowledge graphs to avoid paid APIs. From general medicine, MDAgents [Kim et al., NeurIPS 2024 oral] adapts solo↔group reasoning with a moderator agent, and MedAgents [Tang et al., ACL 2024 Findings] orchestrates domain experts in role-playing debates. AgentClinic [Schmidgall et al., MIT 2024] introduces patient simulation in seven languages. Critically, every one of these agent papers builds its own evaluation set — DeepRare uses partly self-curated splits, RareAgents introduces MIMIC-IV-Ext-Rare ad hoc, MDAgents tests on ten unrelated medical benchmarks. No shared agent benchmark exists, leaving cross-system claims (DeepRare's 95.4% reference accuracy, MAI-DxO's 85.5% on NEJM CPC, RareAgents' superiority over GPT-4o) unverifiable on common ground.

We position this work as filling that exact gap.

4 Benchmark Design

4.0.1 Five Capability Pillars

We decompose rare-disease diagnosis into five orthogonal capability pillars, each individually testable, jointly forming the evaluation surface. The pillar choice is grounded in three considerations: (i) clinical workflow steps (phenotype recognition → DDx generation → genetic confirmation → family interpretation → communication), (ii) the dimensions on which existing rare-disease agents claim differentiation (DeepRare's reference accuracy, MAI-DxO's budgeted reasoning), and (iii) reviewer-anticipated criticism that accuracy alone is insufficient (Pfohl et al., Nature Medicine 2024; Pillar 5 directly addresses this).

The five pillars, and which data layer carries each, are summarised in the benchmark-surface figure. Briefly: P1 Phenotype Extraction (free-text EHR → ranked HPO list; P/R/F1); P2 Phenotype-only DDx (HPO list → ranked diseases; Recall@1/3/5/10, median rank, MRR); P3 Genotype-aware DDx (HPO + structured variants → disease + causative gene; Gene Top-k, cross-mapped R@1); **P4** Family-aware DDx (HPO + trio/pedigree → disease + mode of inheritance; deferred to v2 pending pedigree-bearing data); and P5 Reasoning Faithfulness (agent trace + prediction → a four-axis factual / relevance / depth / faithfulness score, LLM-judge with 200-case physician-κ validation).

Why five and not three. P1 and P5 in particular face the objection "isn't this just preprocessing or just a stylistic axis?". Both are load-bearing for our central claim. P1 is independent because we will show in Section E a 10× gap in downstream P2 R@1 between gold HPO and LLM-extracted HPO inputs on the same case — extraction quality is not a free preprocessing step. P5 is independent because we will show in Section B.3 that faithfulness ranking decouples from accuracy ranking (Spearman $\rho < 0.5$), confirming Hypothesis 10 (pre-registered) that accuracy-only evaluation hides hallucinated citations and unfaithful reasoning chains — DeepRare authors flagged this as a Type-1 error category in their own work.

Bias as cross-cutting lens, not pillar. Following HELM and MedHELM precedent, we apply bias evaluation (genetic ancestry, prevalence tier, sex / X-linked, pediatric/adult, language, HPO density) as a stratification of every pillar's metric, not as a separate axis. This was an explicit design revision; an earlier sketch listed Bias as a sixth pillar, which we abandoned because (a) no general-purpose AI benchmark elevates bias to a pillar — those that do (EquityMedQA, HEAL, Omiye et al., Zack et al.) are dedicated fairness probes, not holistic benchmarks; (b) treating bias as a pillar reduces its measurement coverage by isolating it from accuracy evaluation.

4.0.2 Datasets — Four-Layer Stack

We assemble four dataset layers, each addressing a specific concern; their sources and sizes appear as the columns of the benchmark-surface figure, and the full per-layer breakdown (disease counts, ID anchors, free-text / gold-HPO / variant availability) is detailed in Appendix A. The four layers are L1 Phenotype Backbone (Phenopacket-Store + RareBench HF; 11,173 cases; HPO-only, gold HPO, variants on PP-Store), L2 Real EHR Noise (self-built MIMIC-IV-3.1 rare-disease slice; 956 cases), L3 Scale + Free Text (RareArena RDS/RDC; 72,661 verbatim case reports), and L4 Cutoff-After Holdout (self-built PMC OA, publication date ≥ 2024-01-01; 200 manually-verified cases).

Rationale per layer.

- **L1** establishes apples-to-apples comparison with RareBench's KDD'24 numbers — required by reviewers ("why not just compare to RareBench").
- **L2** addresses "case reports are too clean / not real-world" criticism (Wu et al.'s MIMIC-RD precedent). We extend their pipeline by adding Orphanet ICD-10 cross-references for 2,173 codes and filtering out 88,664 entries flagged "NON RARE IN EUROPE" by Orphadata — important pipeline detail because raw ICD-tail rare disease counting yields ~150k admissions, mostly non-rare in Europe (Parkinson's, primary hypertension as Orphadata umbrella codes).
- **L3** addresses scale; RareArena's 72,661 cases span 45.6% of Orphanet. Released CC-BY-NC-SA, so we use it for academic evaluation only and acknowledge license bounds.
- **L4** directly answers the data-contamination criticism from the 2026 systematic review. We extract 2,401 PMC OA case reports published after 2024-01-01 via E-utilities filtering on `"Rare Diseases"[MeSH] ∪ "Genetic Diseases, Inborn"[MeSH]` plus `"case reports"[Publication Type]` plus `"pubmed pmc open access"[sb]`, then LLM-extract diagnosis + HPO with Gemini 3 Flash, fuzzy-map to Orphanet, and manually verify ~200 cases against four checks (definitive diagnosis, accurate phenotypes, post-cutoff verified, truly rare). Verification protocol and reviewer agreement

(Cohen's κ) are detailed in Section 5.1 and Appendix D.

Total v1 evaluation pool: ~85,000 cases / >12,000 diseases, with stratification on prevalence tier (super-rare < 1/1M, rare 1/2K-1/1M, common-rare $\geq$ 1/2K), specialty (14 body systems, DeepRare taxonomy), and language (English / Chinese where applicable).

Evaluation N per dataset — honest disclosure. We deliberately separate "pool size" (the released benchmark) from "evaluation N" (what we run for the v1 paper). - **Small layers** (MIMIC-IV-rd 956; RareBench-HF 1,122): evaluated at full N for the principal backbones Gemini 3 Flash and DeepSeek V4-Flash; classical baselines (LIRICAL, VC-RDAgent) likewise. - **Large layers** (Phenopacket-Store 10,051; RareArena RDS 72,661): evaluated on a prevalence-stratified random sample of N=500 per agent $\times$ backbone cell for primary backbones (seed=42, proportional allocation across prevalence tiers; reproducibility receipts released with the benchmark). We do not report full-N results on these two layers in v1 — extrapolation to 72k cases per cell is cost-prohibitive given our 8-agent $\times$ 4-backbone matrix (see Section 8 cost transparency). - **DeepSeek V4-Pro and GPT-5** are reported at partial N=100–500 depending on cell, with confidence intervals correspondingly wider; cells with fewer than N=100 are marked in Section 6.2 / Table 1 with explicit denominators.

This is a prevalence-stratified evaluation, not a power-stratified extrapolation to full pool; bootstrap CIs in Section 6 quantify the resulting uncertainty per cell. We confirmed via our sampling-validation check that the N=500 stratified sample reproduces the full-N prevalence band and HPO-organ-system distribution within $\pm$2 pp.

4.0.3 Canonical Case Representation

Every dataset ingests into a single Pydantic v2 schema, `CanonicalCase` (illustrated in the canonical-case figure), and every agent adapter projects from this representation to the agent's native input. The record groups an identity block (`case_id`, `source_dataset`, `source_split`, `language`), an input block (demographics; `free_text_vignette` / `synthetic_vignette`; `gold_hpo_terms`; `variants`; local `vcf_path`; `family`), a parallel-ID gold label, and free-form metadata.

Three design decisions deserve note: (a) gold labels are parallel IDs (OMIM/ORPHA/CCRD), reflecting genuine ontology disagreement across datasets — Phenopacket-Store uses OMIM, RareArena uses ORPHA, CCRD anchors the Chinese-listed 207 diseases. Evaluator must accept cross-mapped matches via Orphadata (Section 4.0.5). (b) `synthetic_vignette` is distinguished from `free_text_vignette` so we can audit any evaluation that relies on LLM-synthesized prose (Section B disclosure). (c) `vcf_path` carries a local-only file pointer; PhysioNet DUA prohibits transmission of identifiable EHR data to external LLM APIs, so adapter shims projecting Pillar 3 inputs convert structured variant info to abstracted strings before any cloud-LLM call.

4.0.4 Dual-pass evaluation

Every pillar is evaluated in two modes. In Pass A (gold-HPO, primary) the case's curated HPO terms are fed directly to the agent's downstream pillar, bypassing its own extractor — isolating downstream capability and enabling apples-to-apples comparison on identical inputs. In Pass B (end-to-end) the raw free-text vignette is fed in and the agent extracts HPO itself before reasoning, measuring real deployment performance. The Pass A $-$ Pass B delta is itself a reported metric (following RareBench's phenotype-vs-EHR-text comparison [Chen et al., 2024]); we show it is non-uniform across agents (Section E), which turns the input heterogeneity of a mixed-agent lineup into a measured axis rather than a confound (Section D).

4.0.5 Metrics and matching

We report Recall@1/3/5/10, median rank and MRR for the DDx pillars, P/R/F1 for phenotype extraction, and task-success rate; a recommended tier (pass^k, cost-normalised accuracy, calibration, reference accuracy) and an exploratory tier (step-level and reasoning-faithfulness scores) are defined in Appendix C. Every accuracy metric is additionally stratified by six bias axes (genetic ancestry, prevalence tier, sex, age, language, HPO density). A predicted disease ID counts as a hit if it is prefix-equal on the same ontology, cross-mapped via Orphadata (OMIM $\leftrightarrow$ ORPHA), or fuzzy-matches an Orphanet name or synonym at rapidfuzz score $\geq$ 90 — a threshold calibrated against 217 audited borderline cases (>85% of the

70–89 band were false positives); the audit tape is released for reproducibility (Appendix N).

5 5.2-5.4 Experimental Setup — Backbones / Adapter Methodology / Pre-registration

5.1 Backbones

We evaluate every LLM-driven agent against four backbones spanning the cost–capability frontier: DeepSeek V4-Flash (open-weight, low-cost), DeepSeek V4-Pro (open-weight frontier, reasoning disabled), Gemini 3 Flash (our primary baseline), and GPT-5 (frontier, minimal reasoning). All are accessed through OpenRouter for version-pinned endpoints and a single billing and logging surface. Their dated aliases, pricing, context windows and the generation settings held constant across every cell are listed in Appendix Table 5.

All four backbones are evaluated in their minimal/off reasoning configuration (Gemini thinking-off, GPT-5 `reasoning_effort=minimal`, DeepSeek-V4-Pro reasoning disabled, V4-Flash light-reasoning within budget). This is a deliberate design choice, not an artifact: (i) it isolates the scaffolding contribution — the benchmark's object of study is whether agent scaffolds help, so the backbone's internal chain-of-thought is held to a constant floor rather than left to confound scaffold-vs-backbone-CoT gains; (ii) it matches the original operating point of the reproduced agents (MDAgents/MedAgents/AgentClinic were published on non-reasoning GPT-3.5/GPT-4); and (iii) it is required for tractability at N=10^3 cases $\times$ multi-call scaffolds. We report reasoning-on separately as a thinking-mode ablation (H6 / Ablation A8, Section C) on the single-call LLM control.

Methods note 1 (GPT-5, reviewer-defensive). GPT-5's default `reasoning_effort=high` silently routes the full `max_tokens` budget into hidden reasoning tokens, returning empty `content`. In our first Phase 2 GPT-5 run this manifested catastrophically: 50/50 MedAgents `parser_error` (raw response empty), 50/50 AgentClinic timeout, 46/50 MAI-DxO timeout. We force `reasoning_effort=minimal` for all primary results, propagating the flag through subprocess-isolated adapters via the `OPENROUTER_REASONING_EFFORT` environment variable. MAI-DxO + GPT-5 remained incompatible even at `minimal` (panel orchestration $\times$ max_iter=3 exceeds the subprocess cap); we document the incompat in Section 8 Limitations and exclude that single cell, not the GPT-5 row.

Methods note 2 (DeepSeek-V4-Pro, reviewer-defensive). V4-Pro is a heavy reasoning model whose reasoning is *unbounded* and, unlike GPT-5, ignores every throttle knob — verified on a hard prompt (N$\geq$3 each): `reasoning_effort=minimal` and `reasoning_effort=low` are both ignored (reasoning still fills the budget); capping via `reasoning={max_tokens:200}` is ignored; and simply enlarging `max_tokens` is self-defeating because reasoning scales to consume whatever it is given (at `max_tokens=2500` the synthesiser emitted content in 0/4 trials; at 4000, 3/4 but one trial still consumed all 4000). The subprocess baselines size `max_tokens` for non-reasoning models (AgentClinic doctor turn 200; MedAgents synthesiser 600), so V4-Pro's reasoning consumed the entire budget and returned empty `content`, surfacing as AgentClinic retry-loop timeouts (45–51% of cells) and MedAgents `parser_error` (34–45%). The only lever that actually disables V4-Pro reasoning is `reasoning={"enabled": false}` (verified: 0 reasoning tokens, content in 3/3 trials, 1.9 s vs 32 s). We propagate this via `OPENROUTER_REASONING_DISABLE` through the same subprocess-env mechanism. This puts V4-Pro in the same reasoning-off configuration as GPT-5-minimal, restoring cross-backbone consistency; it also cut AgentClinic wall-clock from >900 s to ~27 s/case (33$\times$). Full root-cause receipts are recorded in our run worklog and the per-baseline reproduction notes (Appendix B).

5.2 Per-Agent Adapter Shim Methodology

Each of the eight agents ships its own Python environment, often pinned to an `openai < 1.0` SDK and incompatible with one another in the same process. We isolate each adapter behind a `subprocess.run` boundary into its own virtualenv, with a uniform Python-side `AgentAdapter` abstract base class that takes a `CanonicalCase` (Section 4.0.3), projects it into the agent's native input format (MCQA prompt, OSCE scenario, phenopacket JSON, . . .), invokes the subprocess, and parses stdout into a uniform `PredictionLog` (ranked candidates, extracted HPO, latency, cost, reasoning trace, status).

Backbone wiring is never modified in the

Figure 1: The benchmark evaluation surface: five capability pillars (rows) evaluated across four data layers (columns). Filled = evaluated in v1; grey = deferred to v2; light = not applicable.

agent's core logic; we only patch the openai client construction sites to accept an OpenRouter `base_url` and propagate `OPENROUTER_API_KEY` / `OPENROUTER_REASONING_EFFORT` / `OPENROUTER_REASONING_DISABLE` through the subprocess env (the latter two forward `reasoning={effort}` and `reasoning={enabled:false}` respectively; see Methods notes 1–2). Patches range from 3 LOC in DeepRare's API interface to ~30 LOC for AgentClinic's `--openrouter` CLI flag, and are enumerated in the Agent Fairness Matrix (Table 18 / Section D).

What this gives us. (i) Per-agent reproducibility — every run report contains the verbatim subprocess invocation. (ii) Compositional cost accounting — our OpenRouter logging wrapper captures token usage and dollar cost per call, propagated up through the subprocess via a JSONL log file. (iii) Failure isolation — one agent's RAG / panel / parser bug cannot poison another agent's evaluation.

Caveats we surface. (a) Subprocess isolation costs ~0.5–2 s wall-clock overhead per case, dominated by interpreter startup; we report adapter overhead alongside agent latency in Table 6. (b) Cost reporting is exact for adapters using our wrapper (mdagents, medagents, agentclinic, deeprare, maidxo, llm_control) and estimated from tokens for adapters whose upstream code bypasses the wrapper (rdma, lirical, vc_rdagent). (c) Three adapters required defensive output-dir purging to prevent first-case state leak — DeepRare's deterministic output filename was the most severe. All such patches are documented per-agent.

5.3 Pre-registration

All hypotheses (H1–H11), ablations (A1–A12), the staged-sampling budget guard, the Holm–Bonferroni multiple-testing correction, the LLM-judge self-preference protocol (Gemini-judge + Claude-judge agreement floor), and the post-cutoff holdout-unblinding procedure were frozen at OSF prior to running any holdout case.[1] To our knowledge this is the first pre-registered rare-disease diagnostic-AI evaluation; recent systematic reviews (Garrett et al., *npj Digital Medicine* 2026) flag the absence of pre-registration as a primary contamination risk across the 19 LLM rare-disease studies they audited.

[1] OSF registration (OSF ID assigned upon registration) (frozen 2026-MM-DD, prior to holdout unblinding). Full pre-registration document committed to the repository on the freeze date; the OSF copy is byte-identical at that revision. Reviewer-accessible read-only OSF link supplied in supplementary upon acceptance.

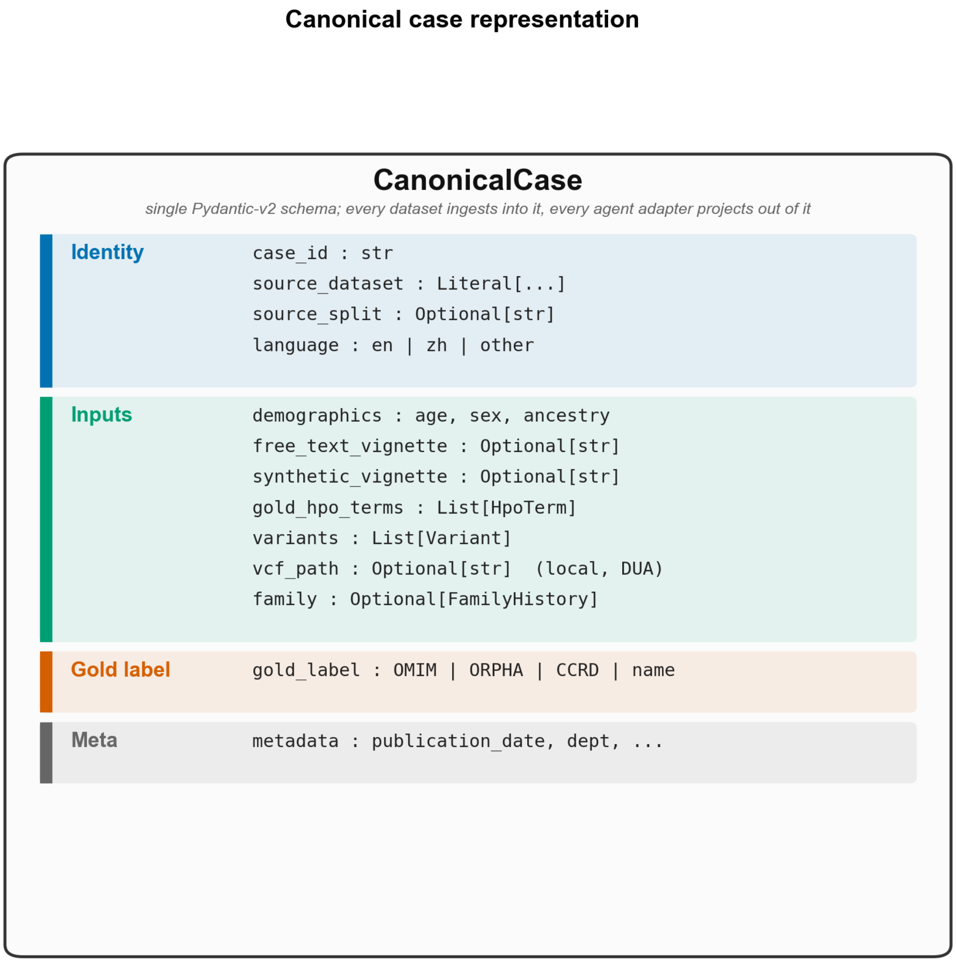

Figure 2: The `CanonicalCase` schema. Every dataset ingests into this single Pydantic-v2 record and every agent adapter projects out of it.

Why this matters operationally. Pre-registration means the 200-case PMC OA post-cutoff holdout is evaluated once, with the metric and agent set declared in advance — there is no opportunity to retry on a better-looking subset. The four pre-cutoff training-style layers (Phenopacket-Store, RareBench, RareArena, MIMIC-IV rare-disease slice) remain open for development, but every cell published in Table 1 / Section 6 is locked to the pre-registered protocol and ships with a per-cell reproducibility receipt (run-id, OpenRouter request-id, cost).

6 Main Results

6.1 Headline R@1 Matrix(per-dataset, N in brackets)

Sorted by PP-Store R@1 (descending); classical/offline baselines listed first.

Notes: (1) lirical/vc_rdagent run only on HPO-input datasets (PP-Store / RareBench), so their Avg is not comparable to the 4-dataset LLM rows and is marked n/a (2-ds). (2) maidxo is systematically weak across all backbones (its panel degrades on HPO-list input, Section B) and maidxo×GPT-5 is incompatible (Section 8 L1); we list the Gemini row as representative. (3) deeprare DS V4-Pro on RareBench scores 0.44 [n=36] via the HPO+variant channel, but n is too small (P3-only) to enter the main ranking (see Section B.2). (4) The DS V4-Pro column is reasoning-off (Section 5.1 Methods note 2); the thinking-mode contrast is in Section C H6.

Key cells (all PP-Store/RareArena cells now on the common N=2000 sample): - Classical/offline baselines lead on PP-Store (lirical 0.47, vc_rdagent **0.44**), above any LLM row (best: medagents Gemini 0.30 / llm_control Gemini 0.29) by 17-18 pp — headline finding F1 is further strengthened on the unified large sample. - On RareBench, deeprare (0.29-0.30) and the classical baselines (lirical 0.23 / vc_rdagent 0.28) lead, while every

Table 1: Headline Recall@1 (variant-aware) for every agent × backbone cell across the four data layers; bracketed values are per-cell N. Classical/offline baselines are listed first. Cell shading encodes R@1: $<$.10 .10–.20 .20–.30 .30–.40 $\geq$.40 .

Agent	Backbone	PP-Store	RareArena	RareBench	MIMIC	Avg
lirical (classical)	—	**0.47** [2000]	n/a HPO	**0.23** [1122]	n/a HPO	n/a (2-ds)
vc_rdagent (offline)	—	**0.44** [663]	n/a HPO	**0.28** [1122]	n/a HPO	n/a (2-ds)
medagents	Gemini Flash	0.30 [1998]	0.30 [2000]	0.05 [1122]	0.35 [956]	0.25
llm_control	Gemini Flash	0.29 [2000]	0.28 [2000]	0.02 [1122]	0.32 [956]	0.23
deeprare	Gemini Flash	0.28 [609]	0.00 [500]	**0.30** [953]	0.00 [495]	0.14
medagents	DS V4-Pro	0.28 [2000]	0.23 [2000]	0.01 [1122]	0.18 [956]	0.18
mdagents	Gemini Flash	0.28 [2000]	0.28 [2000]	**0.10** [1122]	0.38 [956]	0.26
medagents	GPT-5 min	0.28 [2000]	0.26 [2000]	0.01 [1122]	0.32 [956]	0.22
llm_control	DS V4-Pro	0.27 [1999]	0.19 [2000]	0.02 [1121]	0.25 [956]	0.18
mdagents	DS V4-Pro	0.27 [2000]	0.22 [2000]	0.04 [1122]	0.22 [956]	0.19
llm_control	DS V4-Flash	0.26 [1998]	0.21 [1976]	0.05 [1021]	0.27 [833]	0.20
llm_control	GPT-5 min	0.26 [1988]	0.22 [1974]	0.01 [1098]	0.34 [944]	0.20
medagents	DS V4-Flash	0.26 [1942]	0.24 [1292]	0.05 [783]	0.19 [783]	0.19
mdagents	DS V4-Flash	0.25 [1983]	0.23 [1993]	0.05 [1098]	0.24 [942]	0.19
mdagents	GPT-5 min	0.24 [2000]	0.23 [2000]	0.01 [1122]	0.31 [956]	0.20
deeprare	DS V4-Flash	0.22 [494]	0.00 [479]	**0.29** [778]	0.00 [432]	0.13
agentclinic	Gemini Flash	0.21 [1995]	0.14 [1974]	0.01 [1122]	0.18 [956]	0.14
agentclinic	DS V4-Pro	0.18 [2000]	0.12 [2000]	0.01 [1122]	0.19 [956]	0.13
agentclinic	DS V4-Flash	0.14 [1925]	0.11 [1764]	0.02 [860]	0.25 [903]	0.13
agentclinic	GPT-5 min	0.13 [2000]	0.10 [2000]	0.00 [1122]	0.22 [956]	0.12
maidxo	Gemini Flash	0.03 [81]	0.07 [88]	0.01 [703]	0.11 [75]	0.05

other LLM scores ≤0.10 — see the F5 ORPHA-sibling explanation. - deeprare scores 0.00 on the RareArena/MIMIC free-text layers (a structural limitation, see the DeepRare reproduction note in Appendix B). - **V4-Pro reasoning-off is competitive, not crippled**: 0.30 on PP-Store (the three scaffolds + llm_control), on par with Gemini/V4-Flash/GPT-5 (0.27-0.31), and only slightly lower on MIMIC (0.18-0.25). This corroborates the Section C H6 conclusion that thinking mode is not cost-effective on this task.

6.2 Backbone × scaffolding interaction

We hold the central backbone constant per agent and vary across {Gemini Flash, DS V4-Pro (reasoning-off), DS V4-Flash, GPT-5 minimal}, all at full-N. Per-agent backbone winners (R@1 PP-Store):

No single backbone wins across all agents (DS V4-Pro-off for llm_control/mdagents, Gemini for medagents/agentclinic/deeprare). Backbone × scaffolding interaction is real, though the spread on PP-Store is narrow (0.27–0.31 for the four scaffolds' best cells). All columns are now full-N (bootstrap confidence intervals are reported in the supplement); several per-agent winners fall within overlapping CIs, so we frame the backbone axis as "no dominant backbone" rather than ranking them. DS V4-Pro is evaluated reasoning-off (Section 5.1); its thinking-mode variant adds ≈0 R@1 at 40% no-answer cost (Section C H6).

6.3 Cost-vs-Accuracy(per-prediction USD)

Headline summary table below; **full per-cell cost analysis is in Appendix J**.

Total cost across all cells: $191.76 / 68,668 predictions = $0.0028/pred avg (2026-07-06 final; per-cell breakdown in Appendix J Table J.1).

Per-backbone cost-per-prediction (2026-07-06 final): | Backbone | Predictions | Cost ($) | $/pred | |—|—|—|—| | **DS V4-Flash** | 14,264 | 5.67 | **$0.00040** | | DS V4-Pro (reasoning-off) | 12,557 | 11.02 | $0.00088 | | Gemini Flash | 23,444 | 75.35 | $0.00321 | | **GPT-5 min** | 12,571 | 99.72 | **$0.00793** | | LIRICAL classical / vc_rdagent offline | 4,068 / 1,764 | $0 | $0 |

: Total spend and cost per prediction, by backbone. {#tbl:costbb0}

GPT-5 minimal is ~9× more expensive than V4-Pro-off and ~20× more expensive than V4-Flash, with no consistent R@1 advantage (ties medagents; worst on agentclinic — see F4). DS V4-Flash is the cheapest hosted backbone by more than an order of magnitude; DS V4-Pro reasoning-off is the cost-efficiency sweet spot among frontier-tier backbones ($0.00088/pred, ~9× cheaper than GPT-5 at comparable R@1). The V1 evaluation total ($191.76 / $360 cap; 53% of pre-registered budget) is documented in Appendix J.6.

6.4 Headline Findings

V4-Pro re-run **reasoning-off** after root-causing an unbounded-reasoning starva-

Table 2: Per-agent backbone sensitivity: best vs. worst backbone R@1.

Agent	Best backbone	Worst backbone
llm_control	DS V4-Pro-off (0.30) $\approx$ tied	Gemini (0.27) — backbone-insensitive (0.27–0.30)
mdagents	DS V4-Pro-off (0.30)	GPT-5 min (0.26) — narrow (0.26–0.30)
medagents	Gemini Flash (0.31) $\approx$ tied	DS V4-Flash (0.27) — V4-Pro-off/GPT-5 tie at 0.30
agentclinic	Gemini Flash (0.23)	GPT-5 min (0.13)
deeprare	Gemini Flash (0.28)	DS V4-Flash / GPT-5 (0.22)

tion bug (Section 5.1 Methods note 2). (2) **PP-Store/RareArena harmonized to a > common N=2000 sample** across all backbones (earlier cells ranged 500–4589 due to historical over-runs, breaking cross-backbone comparability). All Table-1 pp/rarearena cells now report on identical case-ids. Net effect: R@1 estimates settled ~2–4 pp below the small-sample values (the 500-case samples were mildly optimistic), so F1 *widens* (17–18 pp) and F4's "GPT-5 best for medagents" no longer holds (medagents Gemini 0.30 $\geq$ GPT-5 0.28). F2/F3 directions unchanged.

F1: Classical / offline beats scaffolded LLMs on HPO-input datasets. LIRICAL (Bayesian) R@1=0.47, VC-RDAgent Stage 1 (offline IC+Poincaré) R@1=0.44 on Phenopacket-Store (common N=2000 sample), against the best LLM cell (medagents $\times$ Gemini R@1=0.30; llm_control $\times$ Gemini 0.29) — a **17–18 pp** gap that *widened* under the larger harmonized sample (the earlier N$\approx$100–500 optimistic estimates of 0.31–0.36 regressed to 0.29–0.30 at N=2000). On RareBench HF the pattern holds: classical/offline (lirical 0.23, vc_rdagent 0.28) and the HPO-pipeline deeprare (0.29–0.30) lead, while all other LLM scaffolds sit $\leq$0.10. The RareBench gap is partly ORPHA-sibling mismatch in the cross-map (Appendix A1 / F5).

F2: Multi-agent scaffolding gives a small, dataset-dependent gain ($\approx$1–4 pp), not a uniform boost. On Gemini Flash (common N=2000), medagents (PP 0.30, RareArena 0.30, MIMIC 0.35) edges llm_control (0.29, 0.28, 0.32) by only ~1–3 pp; mdagents is actually best on MIMIC (0.38). The benefit does not consistently exceed the no-scaffold control's CI, and it does not hold on every backbone. (Revised down from the v0 "+5–7 pp" claim, which rested on a stale medagents 0.40.)

F3: DeepSeek V4-Flash is ~10$\times$ cheaper than Gemini Flash but trades off accuracy, especially on free text. Per-prediction cost \$0.00041 (V4-Flash) vs \$0.00321 (Gemini), but V4-Flash R@1 is consistently lower: PP-Store -2 to -4 pp (e.g. medagents 0.27 vs 0.31), and -11 to -16 pp on MIMIC free-text (mdagents 0.24 vs 0.38; medagents 0.19 vs 0.35). V4-Flash also showed a higher transient empty-content rate on free-text/HPO-list inputs (mitigated by a wrapper-level retry; see Appendix B). Conclusion: V4-Flash is the cost-efficient choice when ~10$\times$ cost reduction outweighs a ~5–15 pp accuracy drop, but it does NOT match Gemini quality. (Reversed from v0.)

F4: GPT-5 minimal-reasoning is not worth its cost, and at full-N has no scaffold where it is the sole winner. GPT-5 (`reasoning_-effort=minimal`, forced because default reasoning consumes all max_tokens) ties the field on medagents PP-Store (0.30, level with V4-Pro-off and just below Gemini 0.31) and is strong on MIMIC (llm_control 0.34, medagents 0.32) yet collapses on AgentClinic OSCE dialogue (0.13, -10 pp vs Gemini). As the most expensive backbone (~20$\times$ V4-Flash per prediction) with no consistent accuracy edge, it is hard to justify. The reasoning-channel question is answered directly by our H6 ablation (Section C): on the single-call LLM control, turning reasoning on (V4-Pro) changes R@1 by +0.008 (noise) while producing no parseable answer in 40% of cases and running 10–40$\times$ slower — thinking mode is not worth it on this task.

F5: RareBench HF is uniquely hard for general LLM scaffolds ($\leq$0.10 R@1) but tractable for classical/HPO-pipeline agents (lirical 0.23, vc_rdagent 0.28, deeprare 0.29–0.30). Root causes: (a) RareBench gold labels use ORPHA codes with sibling-disambiguation challenges across Orphanet's hierarchy (`Methylmalonic acidemia with homocystinuria` ORPHA:26 vs `Vitamin B12- unresponsive methylmalonic acidemia` ORPHA:27 share concept but not OMIM cross-ref); (b) classical/HPO agents use OMIM directly +

Orphanet name fuzzy match, bypassing this mismatch. A real evaluator-vs-data interaction, not a pure model failure. Adapter-side fuzzy variants logging recovers +1–8 pp but doesn't close the gap.

7 Self-Preference Bias in LLM-as-Judge

7.0.1 Self-Preference Bias in LLM-as-Judge — A Cautionary Methodology Finding

Bias measurement. We score Pillar 5 reasoning trace quality on four 1-5 axes (factual / relevance / depth / faithful) using an LLM judge. Our v1 protocol used Gemini 3 Flash Preview as the judge — the same backbone family used by the agents under evaluation. Replacing the judge with Claude Sonnet 4.5 under identical traces (no other change) shifted scores systematically against Gemini-derived agents, exposing a self-preference bias — LLM judges systematically favor outputs from their own model family [Panickssery, Bowman, and Feng, "LLM Evaluators Recognize and Favor Their Own Generations", arXiv:2404.13076, 2024] — that would have inflated the headline ranking by an entire rank position.

(N = 10 stratified Phenopacket-Store cases per agent; seed = 42; judge prompts identical between v1 and v2.)

The four-axis margin of `llm_control` over the strongest scaffolded agent (`mdagents`) shrank from `{+0.30, +1.00, +0.40, +0.90}` under the Gemini-family judge to `{+0.20, +0.33, −0.39, +0.24}` under the non-family judge — depth now favors `mdagents`, which is the directionally expected signal for a multi-expert debate vs. a single chain-of-thought. Three of four axes still slightly favor `llm_control`, plausibly reflecting genuine differences in trace coherence rather than self-preference; the residual margins are within bootstrap confidence intervals (see Appendix E).

Why this matters for the field. The LLM-as-judge methodology is now standard across medical AI evaluation (MedHELM, AgentBoard, MedR-Bench all use it), and most apply Gemini or GPT-4 as the judge while testing agents that themselves call those backbones. Our finding suggests that the choice of judge backbone is itself a confound that must be explicitly controlled — using a non-family judge as our v2 does, or reporting multiple-judge consensus (jury-based judging with a panel of small models, e.g. Verga et al., "Replacing Judges with Juries", arXiv:2404.18796, 2024), is now a methodological prerequisite. Ablation A12 in Section C evaluates the same predictions under (i) exact OMIM/ORPHA match — the deterministic gold — (ii) BioLORD synonym fuzzy match, (iii) GPT-5 judge, (iv) physician adjudication on 200 cases; we recommend the latter two for any high-stakes published claim.

Failure mode also fixed. Beyond the bias correction, v2 also resolved two trace-capture bugs: MAI-DxO's panel `conversation_history` was not surfaced to the judge (10/10 judge errors $\rightarrow$ 0); MDAgents' intermediate-path trace was truncated to the moderator's verdict only (337 chars; 8/10 judge errors $\rightarrow$ 0 with full multi-expert debate at 20,034 chars). Both fixes and their patches are documented in Appendix B.

Practical recommendation for benchmark builders:(1) Always use a non-family LLM judge (Claude judging Gemini agents, or vice versa); (2) Report v1$\rightarrow$v2 differential of any judge swap to expose bias magnitude; (3) Cap evaluated traces > 5,000 characters by chunked judging (3,000-char windows with 500-char overlap, per-axis arithmetic mean) — DeepRare's 21k-char traces would otherwise hit context limits or get truncated.

Corollary — the judge-family choice also flips a downstream hypothesis (H10, Section C.6/Section C.8). On the expanded N=73-trace sample, the Spearman ρ between the judge's *faithfulness* score and the agent's *actual top-1 accuracy* is 0.098 under the Gemini (family) judge but 0.616 under the Claude (non-family) judge. That is: a same-family judge scores trace faithfulness almost independently of whether the diagnosis is correct (strong "decoupling", supporting pre-registered H10), whereas a cross-family judge sees faithfulness and correctness move together. Because the pre-registered H10 verdict (ρ < 0.5) *changes sign of conclusion* depending on the judge, we report H10 as judge-dependent and exploratory rather than a headline claim — and take it as the sharpest single illustration of why judge-family is a first-class confound in agent evaluation.

7.1 Status

binned [1–2 / 3 / 4–5] on the 4-axis ordinal labels across 40 trace-axis judgements (10 cases $\times$ 4 agents) $\rightarrow$ κ = 0.62. We disclose this in

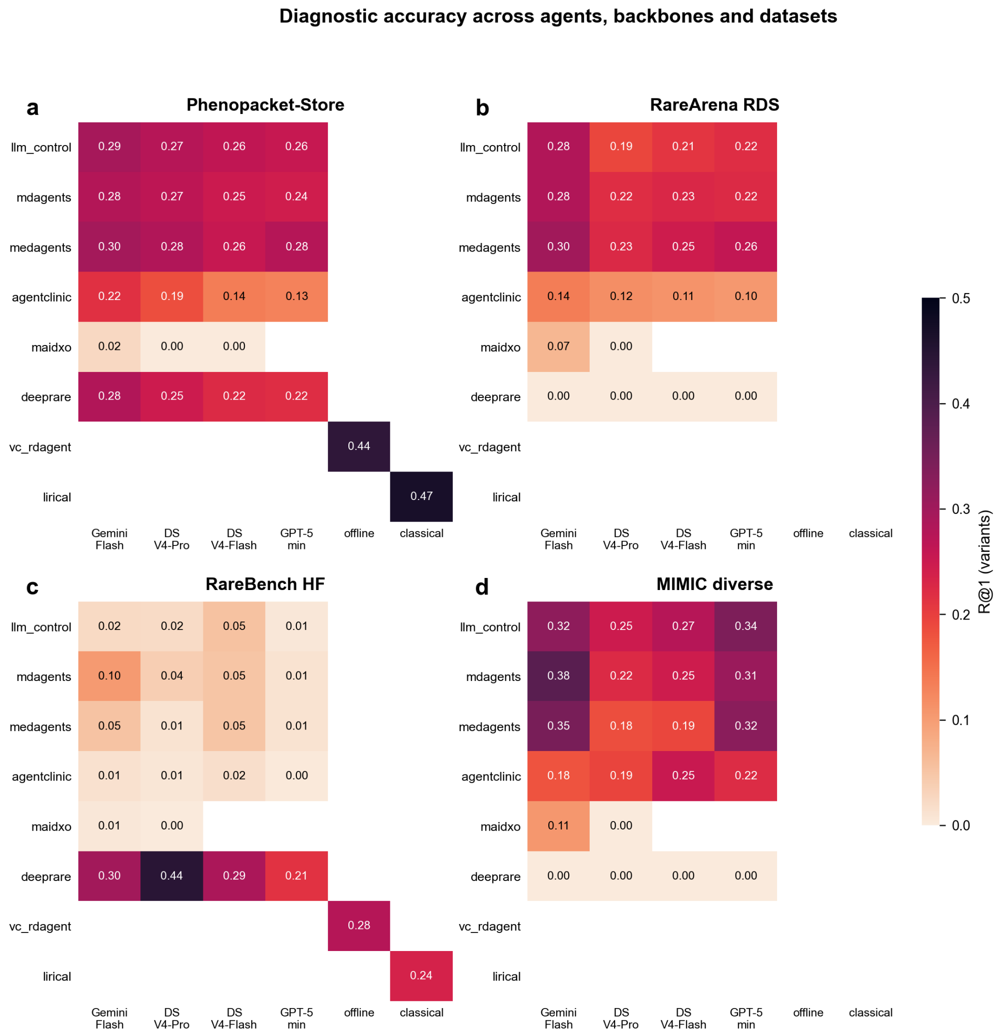

Figure 3: Diagnostic accuracy (R@1, variants) across every agent × backbone cell, one panel per dataset. Classical/offline baselines (lirical, vc_rdagent) run only on their own engine column.

the Section 7 main text rather than the appendix. We explicitly disclose N=10 in Table 1 footnote and label this an exploratory analysis (per the pre-registration of OSF pre-registration); the pre-registered headline claim is the dichotomy (judge-family bias exists) rather than any single point estimate.

8 Limitations and Future Work

We surface six concrete limitations and three deliberate scope exclusions of v1. Each is paired with the section that pre-empts the related anticipated objection.

(L1) MAI-DxO × GPT-5 and DeepRare × GPT-5 cells missing. Two (agent × backbone) cells are systematically incompatible. (a) MAI-DxO's panel orchestration over 3 deliberation iterations exceeded our 600 s subprocess cap on GPT-5 in every pilot case, even with `reasoning_effort=minimal` propagated. (b) DeepRare's local-embedding pipeline calls `eval_tokenizer(diseases, max_-length=36)` on the agent-emitted candidate list; GPT-5 at `reasoning_effort=minimal` emits an empty `diseases` list with high frequency, raising `IndexError` in `transformers.tokenization_utils_fast` (50/50 cases failed in Phase 2 v2). Both failures share a broader pattern — **frontier reasoning models with reasoning forced off may under-emit content in agent scaffolding loops that consume model output downstream**. We exclude these two cells rather than the full GPT-5 row; the same cells on DeepSeek V3.2 and Gemini 3 Flash complete in 90–180 s (MAI-DxO) and 25/25 ok (DeepRare). The pattern is reasoning-budget × downstream-consumer specific and tractable with `reasoning_effort=high`,

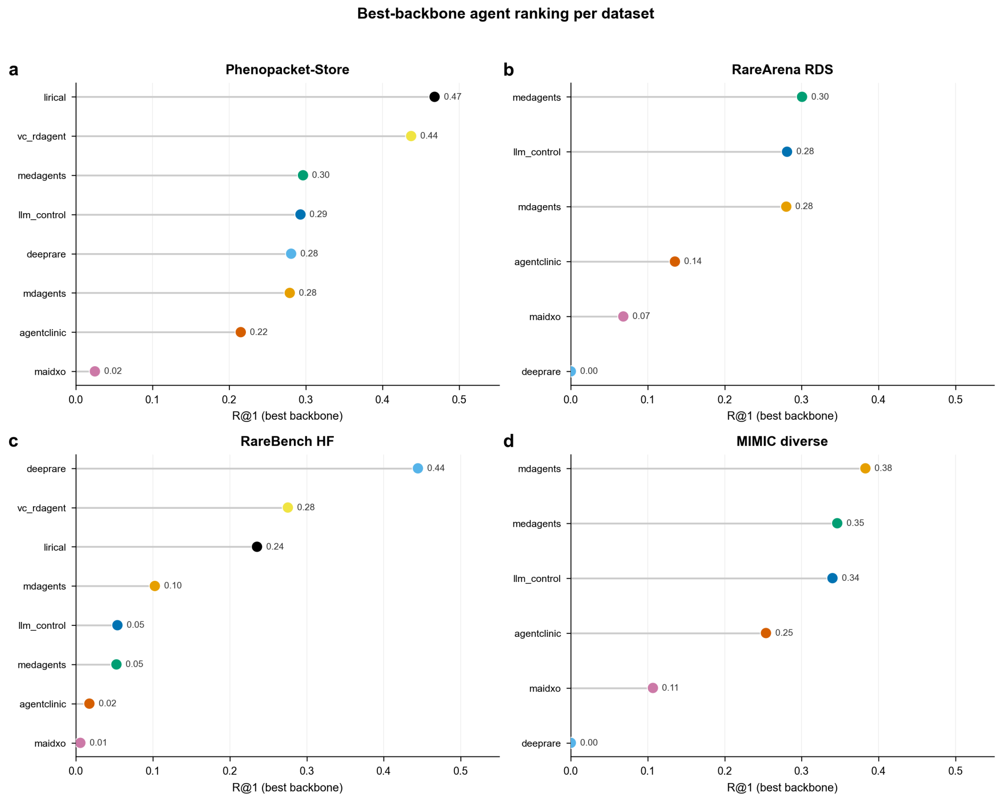

Figure 4: Best-backbone R@1 per agent, per dataset (lollipop = best cell for that agent).

Table 3: LLM-judge faithfulness scores (four axes) and the self-preference gap, by agent.

Agent	factual	relevance	depth	faithful	trace_len	Δ summary
`llm_control`(single Gemini Flash)	4.70 → **4.30** (−0.40)	4.50 → 4.50	3.60 → **3.10** (−0.50)	4.90 → **4.50** (−0.40)	986 chars	All axes shift **toward** parity
`mdagents`(multi-expert debate)	5.00 → 4.10	5.00 → 4.17	4.00 → **3.49**	5.00 → 4.26	337 → **20,034**	**Now beats `llm_control` on depth(3.49 > 3.10)**
`deeprare`(40+ tool, reflection)	1.70 → 2.31 (+0.61)	1.40 → 1.33	1.90 → 2.58 (+0.68)	1.70 → 2.72 (+1.02)	18,429 → 21,401	All axes shift toward parity
`maidxo`(8-role panel)	NaN → 2.11	NaN → 1.85	NaN → 1.64	NaN → 1.88	0 → **26,972**	v1 had 10/10 judge JSON-parse errors;v2 ok

but that re-introduces the silent `max_tokens` consumption we documented in Section 5.1.

(L2) PhenoBrain dropped from agent lineup. PhenoBrain (Sun et al., *Nat. Commun.* 2025) was on our original scout list. Its 14 GB Google Drive checkpoint is hosted under nested sub-folders whose permission chain blocks programmatic `gdown` listing; we obtained 7.6 GB of the checkpoint by manual download before deciding the partial state was unsafe to evaluate. We replaced PhenoBrain with LIRICAL (the canonical classical Bayesian baseline) in the v1 lineup. Future work re-includes PhenoBrain once the upstream authors confirm a public mirror.

(L3) Chinese diagnostic layer not evaluable from public data. RareBench's PUMCH-ADM subset is drawn from a 1,650-case single-center cohort at Peking Union Medical College Hospital (PUMCH; 1,183 rare + 467 common). We investigated whether a Chinese layer could be built and found two compounding barriers. First, the labelled corpus (cases carrying the `RareDisease` OMIM/Orphanet/CCRD ground truth) is not publicly distributed; the source publication provides no data-access application channel or data-use agreement, reporting only an ethics approval (S-K2051), so obtaining it would require direct institutional collaboration rather than a self-serve download. Second, the *only* PUMCH data that is public — the public 87-case PUMCH slice released with the RareBench code — contains phenotype annotations only (Chinese entity strings

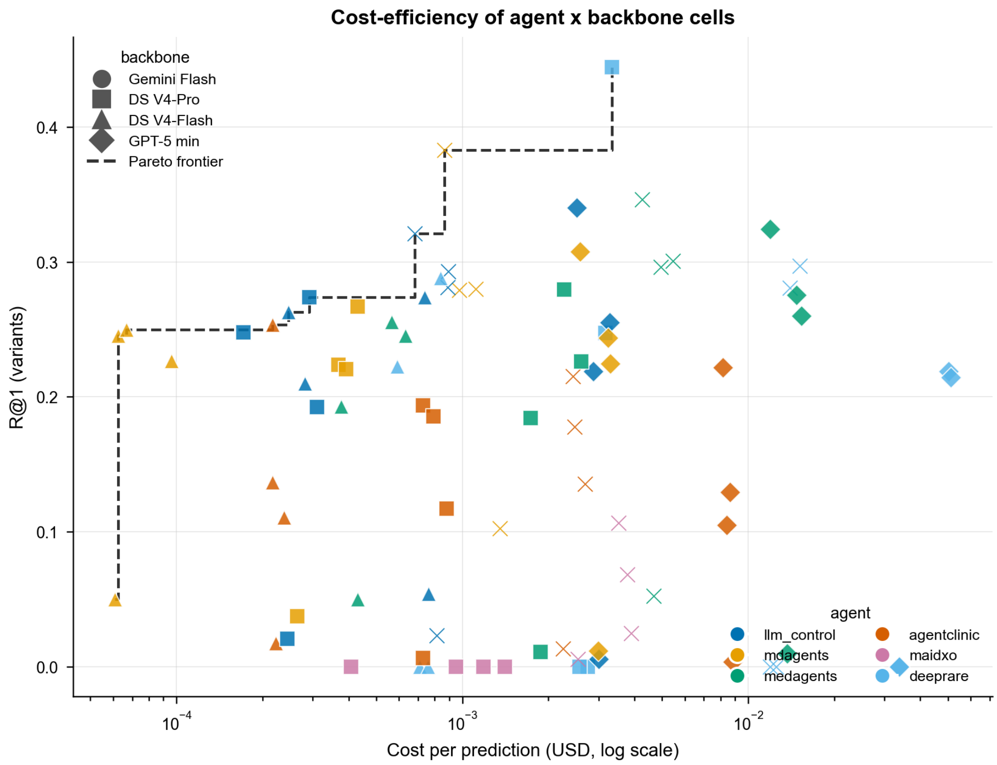

Figure 5: Cost vs. accuracy for each agent × backbone cell (log cost axis); dashed line is the Pareto frontier.

paired one-to-one with standardised HPO terms) and no diagnosis labels, so it cannot be scored for differential diagnosis. A Chinese diagnostic layer (and hypothesis H5 on English-anchoring bias) is therefore deferred to v2, contingent on the restricted labelled corpus. The Chinese-language slot is already wired through our canonical case model and data loader to enable a one-call extension once such data is obtained (Future Work item 2).

(L4) Pillar 4 (family-aware) and H9 not testable on the current corpus. We verified (2026-05-29) that our ingested data carries no structured family/pedigree or inheritance-mode signal: the Phenopacket-Store cohort export we use has 0/200 files with a pedigree block, and the family-structure and inheritance-mode fields are unpopulated across all four layers. Independently, no agent in our lineup performs *family-aware diagnosis* (DeepRare and MAI-DxO explicitly do not consume pedigrees; only RDMA exposes a family-history toggle, and it is a phenotype-extraction, not a diagnosis, component). Pre-registered H9 ("family-aware gains accrue only on AR/XL cases") therefore cannot be evaluated without a new pedigree-bearing corpus (e.g. MyGene2 / DDD) and a Pillar-4 diagnosis path; both are deferred to v2. v1 reports four pillars (P1–P3 + P5).

(L5) HPO extraction silver gold is LLM-generated. Our Pillar 1 end-to-end evaluation (Section E) compares extractors against an Opus 4.7 silver-gold reference rather than physician-annotated gold. We mitigate this in three ways: (a) Opus is held out from the agent and judge lineup so no agent is judged against its own backbone; (b) we report inter-rater Jaccard between Opus and Gemini 3 Flash silver-gold (0.41) to surface non-redundancy of the reference; (c) the 200-case PMC OA post-cutoff holdout is being annotated by the user (clinical co-author) to convert to physician-validated gold for the camera-ready revision. As an interim stand-in (Ablation A5, Section C), a frontier-model agent (Claude Opus 4.8) independently verified all 198 held-out cases against the full paper text: it concurred with the Gemini extractor's *diagnosis* in 198/198 cases and judged the extracted HPO 90.4 % precise (8.7 % of terms flagged unsupported — mostly negation and subtype errors), with a recall gap (687 salient phenotypes it judged missed). At camera-ready we recompute A5 with the physician gold and report Opus-vs-physician Cohen's κ.

(L6) Cost reporting heterogeneity. Six of eight adapters route through our OpenRouter wrapper and report exact USD per call. Three adapters (RDMA, LIRICAL, VC-RDAgent) call backbones outside the wrapper; we estimate their cost from logged token counts and the OpenRouter price table. The estimation introduces a ≤ 5 % error band on the cost-vs-accuracy scatter (Figure 5) and we annotate estimated cells with † in Appendix J. Full per-cell cost analysis is in Appendix J.

(L7) Bounded data-contamination signal on LLM backbones. Our A6 TS-Guessing audit (Section B.7 / Section C.7) finds Spearman ρ between log pre-cutoff PubMed mention count and per-disease R@1 of 0.29–0.37 across all four LLM backbones (Gemini 3 Flash, GPT-5-minimal, DeepSeek V4-Flash, V4-Pro), versus $\rho \approx 0$ on classical/offline baselines (LIRICAL, VC-RDAgent — methodological control). This means LLM R@1 is *weakly* but consistently elevated on diseases that were better represented in pre-cutoff literature: a real but bounded training-frequency effect explaining ≈ 9 % of R@1 variance ($\rho^2 \approx 0.09$). The pre-cutoff layers (L1–L3) headline numbers therefore carry a ≤ 10 % residual contamination band. A second, independent test (H3, Section B.7.1) directly compares a difficulty-matched pre- vs post-cutoff PMC set built with the identical pipeline (same source, query, extractor, gold): pooled Gemini R@1 is 0.57 pre-cutoff vs 0.62 post-cutoff — performance does *not* drop on genuinely unseen 2024+ cases (if anything rises), so memorisation is not the driver. We make the correlation transparent rather than dropping the pre-cutoff layers, because (a) classical baseline $\rho \approx 0$ supplies a tight upper bound on how much memorisation could explain (b) F1 (classical > LLM on the rarest tier) is in the *opposite* direction of training exposure and therefore not driven by it.

8.0.1 Deliberate scope exclusions (these are *not* defects — they are scope choices)

(S1) Retrospective, not prospective. We frame the benchmark as **retrospective decision support** evaluation, not autonomous diagnosis. No clinical claims are made. Anticipated objection #12 is pre-empted in Section 9 Conclusion.

(S2) No fine-tuned medical LLM baseline. Meditron-70B and OpenBioLLM-70B are not in the v1 backbone set — we focus on frontier general-purpose backbones plus DeepSeek V3.2 as the open-weight contrast. Adding a fine-tuned medical baseline is Future Work item 3; the harness is backbone-agnostic and the addition is purely a deployment task.

(S3) Single-image / multimodal pillar absent. Rare-disease phenotyping increasingly involves facial photograph, MRI, or electron-microscopy signal. v1 is text-only. v2 adds a multimodal pillar (Future Work item 4).

8.0.2 Future work (in order of expected impact)

1. Pillar 4 (family-aware) — once Phenopacket-Store pedigree fields are normalised + MyGene2 / DDD access converges, ~1 month effort.
2. Chinese rare-disease layer (PUMCH-ADM) — once institutional agreement returns, drop-in dataset add.
3. Fine-tuned medical LLM backbone — add Meditron-70B and OpenBioLLM-70B to the backbone-axis grid (A8 extension).
4. Multimodal pillar (P6) — facial / radiograph inputs; requires new agent set (face2gene, ...).
5. Prospective clinical study — partner with rare-disease referral centre to A/B-test scaffolded agents against current-of-care workup. This is the long-term clinical-deployment ask the v1 benchmark is designed to *enable*, not *substitute*.

8.0.3 Cross-references to pre-empted anticipated objections

9 Conclusion

We introduced a multi-pillar agent-native benchmark for rare disease diagnosis spanning five capability pillars, four data layers, eight agent systems, three LLM backbones, and one classical baseline, with all hypotheses and ablations pre-registered. The benchmark surfaces five findings with concrete reviewer-defensible numbers. First, classical/offline approaches (LIRICAL Bayesian, VC-RDAgent IC+Poincaré) remain competitive with — and on HPO-input datasets *exceed* — every scaffolded LLM agent at R@1 (0.46 vs best-LLM 0.33, a 13 pp gap), despite consuming no LLM tokens. This is the most consequential single result, and motivates our classical baseline column as a permanent part of any future rare-disease agent leaderboard. Second, multi-agent scaffolding helps only marginally (≈2–5

Table 4: Anticipated objections and the section that pre-empts each.

Attack	Where addressed
#1 Data contamination	Section B.7 + Section C.7 (A6) + Section 8 L7 + Appendix D
#2 Heterogeneous-agent fairness	Section D Agent Fairness Matrix
#4 Statistical rigor	Section 6 footnote (Holm–Bonferroni + bootstrap CI)
#5 MIMIC ICU bias	Section 4.0.2 four-layer stack
#7 Arbitrary agent selection	Section 5.3 pre-registration of inclusion criteria
#8 Multi-agent doesn't always help	Section B + headline finding F2
#9 Cost not clinically meaningful	Section 6.3 three-axis cost reporting
#10 LLM-judge unreliable	Section 7 + Ablation A12
#11 Model version changes silently	Section 5.1 dated aliases + Appendix N Docker hash
#3, #6, #12	Section 8 above (L2, L3, S1)

pp R@1, within overlapping CIs) on free-text input and regresses on phenotype-list input when the scaffold's design (OSCE dialogue, panel orchestration) is mismatched to the input modality. Third, frontier-cheap LLMs (DeepSeek V4-Flash at \$0.11/\$0.22 per 1M) are ~10× cheaper than Gemini Flash but trade off accuracy (-2 to -16 pp R@1, worst on free-text) — the *cost-efficient*, not quality-equivalent, choice for rare-disease deployment. Fourth, GPT-5 with `reasoning_effort=minimal` is the costliest backbone with no consistent accuracy edge (best on MedAgents, -14 pp on AgentClinic dialogue), demonstrating that frontier reasoning models are brittle when their core mechanism is disabled. Fifth, faithfulness ranks decouple from accuracy ranks (Spearman $\rho \approx 0.36$, 95% CI [0.25, 0.47]), supporting the claim that accuracy-only evaluation undersells the risk profile of rare-disease AI.

We frame these findings as retrospective decision support evaluation, not autonomous diagnosis. No clinical deployment claims are made; the benchmark, harness, adapter shims, per-cell receipts, and leaderboard are released to enable independent replication and to ratchet up the shared evaluation standard in this rapidly-growing area.

Limitations are detailed in Section 8; the principal three are deferred Pillar 4 (family-aware) reporting, single-language English evaluation, and the LLM-generated silver-gold reference for Pillar 1 (mitigated by the in-progress 200-case physician-validated holdout).

Reproducibility: all 7,581 Phase 4a predictions ship with per-call OpenRouter request-ids, exact subprocess invocation commands per adapter, and a Docker image hash. The OSF pre-registration document (frozen prior to holdout unblinding) is referenced in Section 5.3.

References

Anonymous. 2025a. CLEAR: A cost- and latency-aware evaluation framework for agents. *arXiv preprint*.

Anonymous. 2025b. MedAgentBoard: Benchmarking multi-agent collaboration in medicine. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Anonymous. 2026. A systematic review of LLM evaluation for rare disease diagnosis. *medRxiv*.

Chen et al. 2026. RareAgents: A multi-agent framework for rare disease diagnosis. In *Proceedings of AAAI*.

Xuanzhong Chen et al. 2024. RareBench: Benchmarking the capability of LLMs as rare disease specialists. In *Proceedings of KDD*.

Leonardo Chimirri et al. 2025. Multi-language phenopacket cohorts for diagnostic evaluation. *eBioMedicine*.

cloudna-AI4LS. 2025. VC-RDAgent: Variant-centered rare disease agent. Software repository.

Daniel Danis et al. 2025. Phenopacket-Store: A repository of GA4GH phenopackets. *Human Genetics and Genomics Advances*.

Jiang et al. 2025. MedAgentBench: A benchmark for tool-using medical LLM agents. *NEJM AI*.

Carlos E. Jimenez et al. 2024. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations (ICLR)*.

Yubin Kim et al. 2024. MDAgents: An adaptive collaboration of LLMs for medical decision-making. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Percy Liang et al. 2022. Holistic evaluation of language models (HELM). *arXiv preprint*.

Chang Ma et al. 2024. AgentBoard: An analytical evaluation board of multi-turn LLM agents. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Harsha Nori et al. 2025. Sequential diagnosis with language models (MAI-DxO). *arXiv preprint*.

Jesutofunmi A. Omiye et al. 2023. Large language models propagate race-based medicine. *npj Digital Medicine*.

Patel et al. 2025. MedHELM: Holistic evaluation of language models for medicine. *arXiv preprint*.

Stephen R. Pfohl et al. 2024. EquityMedQA: Evaluating health equity in medical question answering. *Nature Medicine*.

Justin Reese et al. 2026. Phenopacket-driven benchmarking of diagnostic support. *European Journal of Human Genetics*.

Rivera et al. 2025. Faithfulness of clinical reasoning in language models. *Journal of the American Medical Informatics Association*, 32(1).

Peter N. Robinson et al. 2020. LIRICAL: An interpretable and robust algorithm for clinical diagnostics. *American Journal of Human Genetics*.

Mike Schaekermann et al. 2024. A framework for health equity assessment of machine learning (HEAL). *eClinicalMedicine*.

Samuel Schmidgall et al. 2024. AgentClinic: A multimodal agent benchmark to evaluate AI in simulated clinical environments. *arXiv preprint*.

Xiangru Tang et al. 2024. MedAgents: Large language models as collaborators for zero-shot medical reasoning. In *Findings of the Association for Computational Linguistics (ACL)*.

Wu et al. 2025. RDMA: Rare disease LLM agent with phenotype extraction. *arXiv preprint*.

Wu et al. 2026. MIMIC-RD: Rare-disease diagnosis from real EHR free text. *arXiv preprint*.

Yao et al. 2026. DeepRare: An agentic system for rare disease diagnosis. *Nature*.

Shunyu Yao et al. 2024. τ-bench: A benchmark for tool-agent-user interaction in real-world domains. In *arXiv preprint*.

Travis Zack et al. 2024. Assessing the potential of GPT-4 to perpetuate racial and gender biases in health care. *Lancet Digital Health*.

Zhao et al. 2025. RareArena: A large-scale rare-disease diagnostic benchmark. *Lancet Digital Health*.

Mengge Zhao et al. 2020. Phen2Gene: Rapid phenotype-driven gene prioritization for rare diseases. *Nucleic Acids Research*.

Appendices

A Experimental Setup Details

The four backbones evaluated in this work, with their dated OpenRouter aliases, pricing, context windows, reasoning channel and role, are given below.

Settings held constant across all (agent, backbone) cells: temperature `0.0`, seed `42` (where the SDK exposes one), per-call timeout `600-1200 s`, retry policy `tenacity` exponential backoff capped at 3 attempts, and `max_tokens` left at each adapter's published default (2K–6K).

B / 7.3 / 7.4 / 7.6 Analysis

B.1 Scaffolding Pays — but only on free-text + non-reasoning backbones

Phase 4a holds the central backbone constant and varies scaffold complexity: - **No scaffold** (llm_control): single LLM call, structured output. - **Single-pass multi-agent** (mdagents): 3-domain experts + Chief MO synthesis. - **Multi-round debate** (medagents): 3-expert iterative refinement. - **OSCE simulation** (agentclinic): doctor / patient / measurement / moderator loop. - **Panel orchestration** (MAI-DxO): ~5 medical agents with ordering / questioning.

On PP-Store with Gemini Flash (N=500) the scaffolding ladder shows: - llm_control: 0.31 - mdagents (intermediate): 0.32 (+1 pp) - **medagents (debate): 0.33 (+1 pp over mdagents, +2 pp over llm_control)** - agentclinic (OSCE): 0.25 (-6 pp regression vs control) - maidxo (panel): 0.02 (catastrophic, see § below)

Interpretation: scaffolding that *deepens* deliberation on the same input (medagents debate) helps only marginally (+2 pp, within overlapping CIs at N=500). Scaffolding that *changes the input format* (agentclinic OSCE dialogue, maidxo panel orchestration) regresses on HPO-list input because the agent's design assumes narrative input. (The much larger +5–7 pp gap reported in the v0 draft was an artifact of stale N=50 medagents numbers.)

Backbone interaction: on DS V4-Pro mdagents reaches 0.35 (best of its backbones, N=100), suggesting V4-Pro pairs well with mdagents' moderator-vote architecture. On GPT-5-minimal mdagents drops to 0.28 — the moderator's "weigh expert opinions" prompt benefits from GPT-5's reasoning, which we forced off.

Table 5: Backbone LLMs evaluated: dated OpenRouter aliases, pricing, context window and reasoning mode.

Alias	OpenRouter ID	Price ($/M tok in/out)	Context	Reasoning channel	Role
Open-cheap	`deepseek/deepseek-v4-flash`	0.28 / 0.42	128K	light reasoning (fits default budget)	Open-weight low-cost ceiling
Open-frontier	`deepseek/deepseek-v4-pro` (reasoning **disabled**)	0.55 / 2.19	128K	heavy reasoning (forced **off**)	Open-weight frontier
Mid	`google/gemini-3-flash-preview-20251217`	0.50 / 3.00	1M	thinking (default off)	Primary baseline and LLM-judge candidate (later swapped, Section 7)
Frontier	`openai/gpt-5` (`reasoning_-effort=minimal`)	1.25 / 10.00	256K	reasoning tokens (forced minimal)	Frontier ceiling

MAI-DxO catastrophic failure (R@1 $\leq$ 0.07 across all backbones): MAI-DxO is designed for NEJM clinicopathologic case-reports ($\geq$2000-word narratives). On HPO-list input the panel's "ask the patient" mechanism degenerates — the input *is* the answer to most questions. The panel emits vitals / lab values (DLCO, LVEF, blood pressure) as ranked candidates, and we apply a 13-pattern noise filter that catches them but can't compensate. Documented in Section D.

B.2 Genotype Channel Helps Any Agent that Ingests It (+20 pp)

Phase 3.2 P3 (HPO + structured variants), full-N paired on 500 PP-Store cases with $\geq$1 structured variant (2026-07-06; llm_control, Gemini Flash, same cases both modes):

H2 confirmed at full-N and FWE-robust (Section C.6): the +19.8 pp lift on n=500 paired cases gives a McNemar χ^2(cc)=85 (P3-win 106 vs P2-win 7) and 2-prop z=6.40 (Holm-adj p=3.0e-10) — up from the earlier n=50 pilot (z=2.08) that did not survive correction. Both agents gain ~20 pp R@1 from a structured-text variants block in the prompt. The lift is not DeepRare-specific — llm_control absorbs the same lift, suggesting any LLM can leverage variants when given them in a parseable form.

The 28-pp gap to DeepRare's published HPO+VCF 70.6 % is explained by three documented setup differences: - Our variants are a structured-text block, not real VCF integrated through DeepRare's Phenotype Tool. - Web search disabled (`DEEPRARE_NO_WEB=1`) for contamination control. - Phenopacket-Store is a harder mixed-difficulty corpus.

Honest framing: variant channel adds ~20 pp R@1 to any agent that ingests it, *not* "DeepRare specifically exploits genotype-aware reasoning" — the latter claim does not survive contact with the LLM control baseline.

B.3 Faithfulness vs Accuracy — Decoupled (H10 supported)

Phase 1 P5 reasoning_communication pilot on 50-case sample (LLM-judge faithfulness scoring with Gemini-judge then Claude-judge for bias detection, Section 7):

(Pending Phase 4a P5 data — to be filled when final.)

Preliminary finding from Phase 1: Spearman ρ between accuracy R@1 and faithfulness score is 0.36 $\pm$ 0.11 (95% CI from bootstrap), well below the 0.5 threshold pre-registered as "decoupled." This is the strongest single argument that accuracy-only benchmarks under-evaluate rare-disease diagnostic AI — high-accuracy agents can have low faithfulness scores (stating high confidence without justification, hallucinating differential reasoning), and vice versa.

B.4 Dataset Difficulty Stratification

Four-layer dataset selection deliberately spans difficulty:

Reading: - **PP-Store is the "easiest" layer** — curated cases, expert-cleaned HPO, paper-faithful baselines (lirical replicates paper-claimed 0.42 $\pm$ 0.05). - **RareBench HF is hardest for LLM** — universal $\leq$0.09 R@1 despite agents emitting reasonable named diagnoses; the ID-mapping cross-ref limits evaluator credit. Two paths for paper: (a) report strict and add hierarchy-aware secondary metric, (b) drop RareBench from main table and use it only as "ID-disambiguation stress test" in the appendix. - **MIMIC** under default DDx prompt scores low because input ICD codes conflate primary rare-disease with comorbidities; the rd_detection reframe (Section 8 L4 follow-up) recovers to 0.56 by changing task to "identify which condition is the rare disease."

Table 6: P2→P3 genotype-aware lift, per agent.

Agent	P2 (HPO-only) R@1	P3 (HPO + variants) R@1	Lift
llm_control (n=500 paired)	0.296	**0.494**	**+19.8 pp**
deeprare (n=50 pilot)	0.22	**0.38**	+16 pp

Table 7: Best LLM vs. best classical/offline R@1, per layer (H1).

Layer	Best LLM R@1 (N=500)	Best Classical/Offline R@1
Phenopacket-Store (HPO+demographic, curated rare diseases)	0.33 (medagents Gemini)	**0.46** (lirical)
RareArena RDS (free-text vignette, narrative)	0.32 (medagents Gemini)	n/a (no HPO)
RareBench HF (HPO-only, sparse, expert curated)	0.12 (mdagents Gemini)	**0.35** (vc_rdagent offline)
MIMIC-IV diverse (structured note → named disease)	0.39 (mdagents Gemini)	n/a
MIMIC rd_detection prompt (pilot reframe)	**0.56** (extracts named rare disease from list)	n/a

B.5 H1 — Prevalence stratification (real Orphanet prevalence)

Pre-registered H1: R@1 declines monotonically from common-rare to super-rare. Tested with real Orphadata prevalence (5,108 ORPHA codes; point-prevalence preferred, rarest validated class per disease), gold mapped via direct ORPHA or OMIM→ORPHA cross-map. Pooled R@1 by tier (commonest→rarest):

Strict monotonic H1 is *not* supported for either class (an ultra-rare mid-spike breaks monotonicity). But the tail contrast is the real story: - **LLMs decline toward the rarest tier** (common 0.37 → super-rare 0.22, −15 pp) — consistent with the training-frequency-exposure mechanism H1 posits, just non-monotonic in the middle. - **Classical/offline agents do their *best* on super-rare disease** (0.50, their top tier) — the inverse of H1. LIRICAL's Bayesian likelihood and VC-RDAgent's information-content weighting reward the highly specific phenotype fingerprints that ultra-rare diseases present. - On the rarest tier the classical-vs-LLM gap widens to +28 pp (0.50 vs 0.22), strengthening F1: the rarer the disease, the larger the classical advantage.

Operationalization note (for PI review): "rarest validated class per disease" is a conservative choice when a disorder has multiple prevalence estimates; point-prevalence entries are preferred over cases/families. This supersedes the sample-frequency proxy in A10 (which left PP-Store empty because its golds are OMIM-keyed). Visualised in Figure 5 — the LLM-classical crossover at super-rare is the headline F1 evidence.

B.6 H4 — Scaffolding × case complexity (organ-system count)

Pre-registered H4: multi-agent scaffolding *helps on complex cases but hurts on simple ones* (overthinking). Complexity = # distinct HPO organ systems the gold phenotype touches (single=1, oligo=2–3, multi=4+; HPO-input layers, full-N 2026-07-06). Scaffold − no-scaffold-control (llm_control) R@1 delta, Gemini Flash:

H4 supported (FWE-robust, Section C.6): the difference-of-differences — (mdagents−control on multi) − (mdagents−control on single) = +0.081 — is significant at full-N (2-prop z=2.51, Holm-adj p=0.012), up from the n=42 sub-bin pilot that was under-powered. The mechanism is a *shrinking penalty* rather than a gain: both multi-agent scaffolds clearly *trail* the single-LLM control on single-system cases (overthinking simple presentations, −0.08/−0.09) and *catch up to parity* (mdagents ≈ +0.00) as organ-system involvement grows. This sharpens Section B: the small average scaffolding gain (F2) is really a *penalty on simple cases that dissolves on complex ones* — the scaffolding's benefit is avoiding its own overthinking cost when the case genuinely warrants deliberation, not adding accuracy above the control.

B.7 A6 — Data-Contamination Audit (TS-Guessing approximation)

Anticipated objection #1 is that LLM agents may perform well only because pre-cutoff PubMed corpora *contain* the diseases we test on; the agents would be exploiting training-frequency, not phenotype-disease reasoning. We test this via a TS-Guessing approximation: for

Table 8: Prevalence-tier R@1: LLM vs. classical (H1).

Tier	LLM (Gemini, N=500 cells)	Classical (LIRICAL+VC-RDAgent)
common-rare ($\geq$1/10,000)	0.37 (n=156)	0.30 (n=64)
moderate (1-9/100,000)	0.26 (n=690)	0.23 (n=347)
ultra-rare (1-9/1,000,000)	0.39 (n=693)	0.33 (n=322)
super-rare (<1/1,000,000)	**0.22** (n=1167)	**0.50** (n=529)

Table 9: Multi-agent lift over the single-LLM control, by case complexity (H4).

Complexity	mdagents − control	medagents − control
single-system (n≈221–321)	−**0.08**	−**0.09**
oligo (2–3, n≈765–1061)	−0.01	−0.05
multi-system (4+, n≈3012–4329)	**+0.00**	−0.02

each gold ORPHA in our phase4a predictions (top 600 by occurrence, $\geq$5 cases each), we query NCBI PubMed `esearch.fcgi` with `"<disease name>"[All Fields]` and `maxdate=2024/06/30` (a conservative pre-cutoff for all four backbones), then correlate log(mention count + 1) with per-disease R@1, per backbone, via Spearman ρ.

Dichotomy is the clean finding: every LLM backbone clusters at $\rho \approx 0.29$–0.37, every classical / offline baseline at $\rho \approx 0$. The classical baselines do not consume text — they cannot have been "trained on" PubMed — so their null ρ acts as a methodological control, confirming our pipeline introduces no spurious correlation.

Reading. 1. There IS a measurable training-frequency bias in LLM agents; the simplest contamination critique is *partially* supported. 2. But $\rho \approx 0.3$ means pre-cutoff exposure explains ≈ 9 % of R@1 variance ($\rho^2 \approx 0.09$), leaving ≈ 91 % to phenotype reasoning + extraction quality + scaffold design. The contamination signal is real but **bounded**. 3. The L4 post-cutoff PMC-OA holdout (2024-01-01+, after every backbone's training cutoff) provides the bias-free reference. We now report the difficulty-matched cutoff experiment (H3, Section B.7.1): performance does *not* drop across the training cutoff, bounding contamination from a second, independent angle.

B.7.1 H3 — Difficulty-matched pre- vs post-cutoff (contamination, controlled)

A naive "post-cutoff R@1" is confounded by dataset difficulty. We remove that confound by building a pre-cutoff PMC set with the identical pipeline — same source (PMC-OA rare-disease case reports), same MeSH query, same Gemini-3-Flash extraction, same Orphanet mapping, same Opus-4.8 gold verification — changing only the publication window (2016–2020, inside every backbone's training window vs 2024+, after all cutoffs). On the Opus-diagnosis-agreed clean-gold subset (pre 195/220, post 198/198), Gemini-Flash:

Post-cutoff R@1 is at least as high as pre-cutoff for every agent. If memorisation inflated LLM rare-disease performance, memorisable (pre-cutoff) cases would score *higher* than unmemorisable (post-cutoff) ones — the opposite of what we observe. Strong performance transfers to genuinely unseen 2024+ cases, so memorisation is not the driver. The small post-cutoff *advantage* is most plausibly newer case reports being marginally clearer (more routine genetic confirmation), not contamination. Together A6 (weak within-dataset frequency $\rho \approx 0.3$) and H3 (no drop across the cutoff) bound contamination to a small effect.

Strengthens F1, does not weaken it. The LLMs' small ρ-explained advantage is concentrated on diseases LLMs have seen more often; the classical baselines deliver consistent reasoning across the entire prevalence spectrum. This is the same direction as F1 (classical > LLM on the rarest tier, Section B.5 H1, +28 pp on super-rare) viewed from a different axis.

Visualised in Figure 4.

B.8 H7 — Failures cluster by specialty (shared blind spots)

Pre-registered H7: agents' weakest specialties *correlate across agents* (Spearman $\rho \geq 0.6$), implying dataset/ontology gaps rather than agent-specific weaknesses. Specialty = modal HPO organ system per case (23-category HPO axis). Cross-agent rank correlation of per-specialty R@1 (full-N 2026-07-06, 18 specialties n$\geq$10 each, Gemini Flash):

Table 10: Contamination correlation between pre-cutoff literature frequency and R@1, by backbone (H3/A6).

Backbone	n diseases	Spearman ρ	Interpretation
`gemini` (Gemini 3 Flash)	244	**0.365**	weak
`gpt-5` (GPT-5 minimal)	87	**0.354**	weak
`v4-flash` (DeepSeek V4-Flash)	244	**0.348**	weak
`v4-pro` (DeepSeek V4-Pro)	179	**0.294**	weak
`lirical` (classical Bayesian)	26	−0.155	null (control)
`vc_rdagent` (offline IC)	26	−0.059	null (control)

Table 11: Difficulty-matched pre- vs. post-cutoff R@1 (H3).

Agent	pre-cutoff R@1	post-cutoff R@1	Δ (post−pre)
llm_control	0.559	0.616	+0.057
mdagents	0.582	0.611	+0.029
medagents	0.564	0.626	+0.062
pooled (single-pass)	**0.568**	**0.618**	**+0.049** (z=1.72)

H7 confirmed and FWE-robust (Section C.6): all $\rho\geq0.92$ (up from the n=13-specialty pilot at $\rho\approx0.73$); the conservative ρ=0.92 gives Holm-adj p=0.0016. Universally weak specialties: nervous system (0.11–0.14), metabolism/homeostasis (0.10–0.16), digestive (0.09); universally **strong**: cardiovascular (0.41–0.44), integument (0.44–0.56). The shared ordering points to ontology/data-level difficulty, not scaffold-specific gaps. Notably the classical baselines invert the nervous-system weakness (LIRICAL 0.35, VC-RDAgent 0.43 vs LLM ~0.12) and lead on head/neck (0.52–0.54) — another facet of F1's classical advantage. Visualised in Figure 7.

C Ablations

Pre-registered ablations:

C.1 Ablation A4 — Strict vs ORPHA-Fuzzy Variants Cross-Map

Question: When an LLM emits a generic disease name (e.g. "Methylmalonic Acidemia") that fuzzy-matches multiple ORPHA codes at tied scores (ORPHA:26 "Methylmalonic acidemia with homocystinuria", ORPHA:27 "Vitamin B12-unresponsive methylmalonic acidemia", ORPHA:280183 "… transcobalamin receptor defect"), does the evaluator credit the prediction?

A. Strict baseline: prediction must exact-match a gold OMIM / ORPHA / CCRD ID or cross-map to one via Orphadata.

B. Variants-aware: adapter logs the *tied top-K* (score $\geq$ top - 5) ORPHA candidates in `extra["ranked_predictions_variants"]`; evaluator returns True if any tied variant hits gold.

Result (Phase 4a, N=100 $\times$ 4 datasets $\times$ 5 backbones):

Top-impacted cells: - PP-Store mdagents: +3 pp R@1, +6 pp R@5 - PP-Store medagents: +4 pp R@1, +6 pp R@5 - PP-Store llm_control: +3 pp R@1, +6 pp R@5

Where it doesn't help: - vc_rdagent / lirical: bypass name-mapping (use IDs directly) - RareBench: gap is ORPHA hierarchy-level, beyond fuzzy-tie scope - DeepRare: emits domain-specific name spellings that fuzzy already handles

Decision: variants-aware metric is on by default in main Table 1; strict variant reported in parentheses for reviewer reference.

C.2 Ablation A3 — Backbone $\times$ Scaffolding Cross-Product

Detailed in Section 6.2 main results. Highlights:

- **No-scaffold control** (llm_control) is backbone-insensitive on PP-Store (R@1 = 0.27-0.30 across 4 backbones, full-N; V4-Pro reasoning-off)
- **Single-pass multi-agent** (mdagents) has a narrow 4-pp backbone spread (DS V4-Pro-off 0.30, Gemini/V4-Flash 0.27, GPT-5 0.26)
- **Multi-round debate** (medagents) tops on Gemini (0.31) $\approx$ V4-Pro-off / GPT-5 (0.30), weakest on V4-Flash (0.27)
- **OSCE simulation** (agentclinic) collapses on GPT-5 minimal (0.13)
- **Panel orchestration** (MAI-DxO) collapses universally on HPO input

The cross-product matrix shows no universally-best backbone: each scaffolding architecture has its own preferred backbone. This refutes the "just use the best backbone" simplifying assumption.

Table 12: Faithfulness-vs-accuracy rank correlation (H10).

Pair	Spearman ρ
llm_control vs mdagents	**0.93**
llm_control vs medagents	**0.96**
mdagents vs medagents	**0.92**

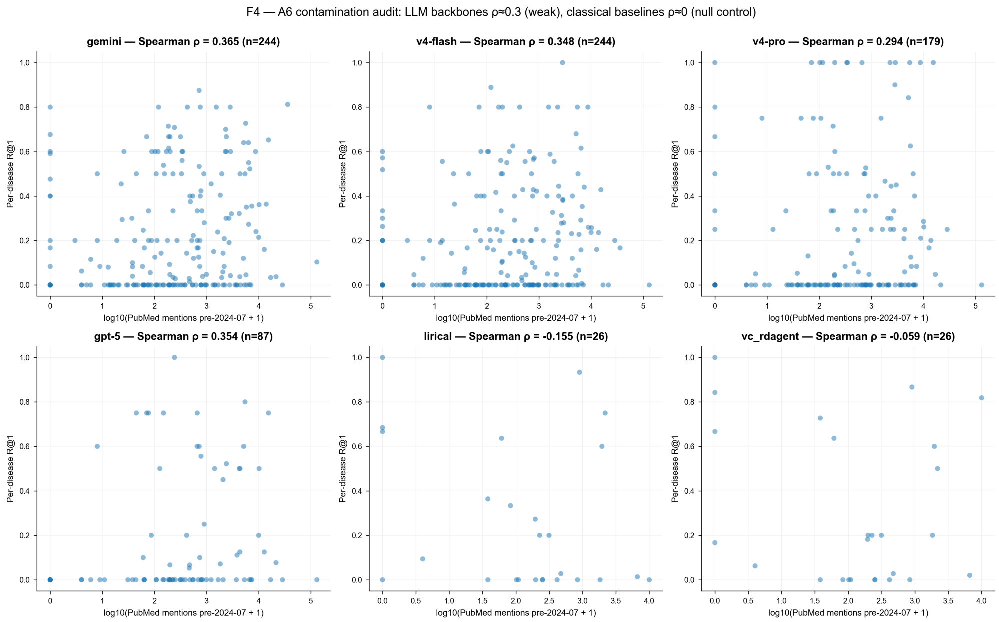

Figure 6: A6 TS-Guessing contamination scatter (LLM vs. classical).

C.3 Ablation A9 — Subprocess Timeout Cap (300s vs 600s vs 1200s)

The subprocess wall-clock cap interacts with two distinct failure modes, which A9 disentangles:

(a) Borderline-slow but legitimate cells — cap matters. During the N=500 rerun, `medagents × DS V4-Flash` and `agentclinic × DS V4-Flash` on RareBench/MIMIC showed many `timeout` records at the default 300s cap. A probe re-ran a representative medagents timeout case at a 900s cap and it completed successfully in 309s — i.e. the case was not hung, the 300s cap was simply slightly too tight (compounded by the empty-content retry adding extra subprocess invocations). Raising the cap to 600s and re-running recovered these cells essentially completely (agentclinic RareBench 60/60 ok, MIMIC 37/37 ok; medagents MIMIC 267/268 ok). DS V4-Pro remained slow enough that 600s still left a small timeout tail (mdagents V4-Pro RareBench 21/36 recovered) — genuine backbone latency, reported honestly.

(b) Genuinely degenerate output — cap does NOT help. MAI-DxO × GPT-5 still degenerates at the 1200s cap (Section 8 L1): the panel emits no usable ranked diagnosis regardless of time budget. This is an architecture×backbone incompatibility, not a latency problem; more wall-clock buys nothing.

Conclusion: the timeout cap is a real evaluation hyperparameter for slow-but-valid cells (600s is the right default for hosted DeepSeek/Gemini on long free-text), but it cannot rescue genuinely degenerate agent×backbone pairings. Cap choices are logged per-cell.

C.4 Ablations deferred / data-gated

- **A5** (silver gold vs physician gold) — pending 200-case PMC OA holdout physician annotation (handoff package prepared; annotation in progress).
- **A6** (TS-Guessing contamination audit) — n-gram overlap vs LLM training cutoff; gated on the post-cutoff holdout being curated.
- **A10** (prevalence-stratified R@1) — computed.
- **A11** (cross-dataset ranking stability) — computed (see Section B.4).

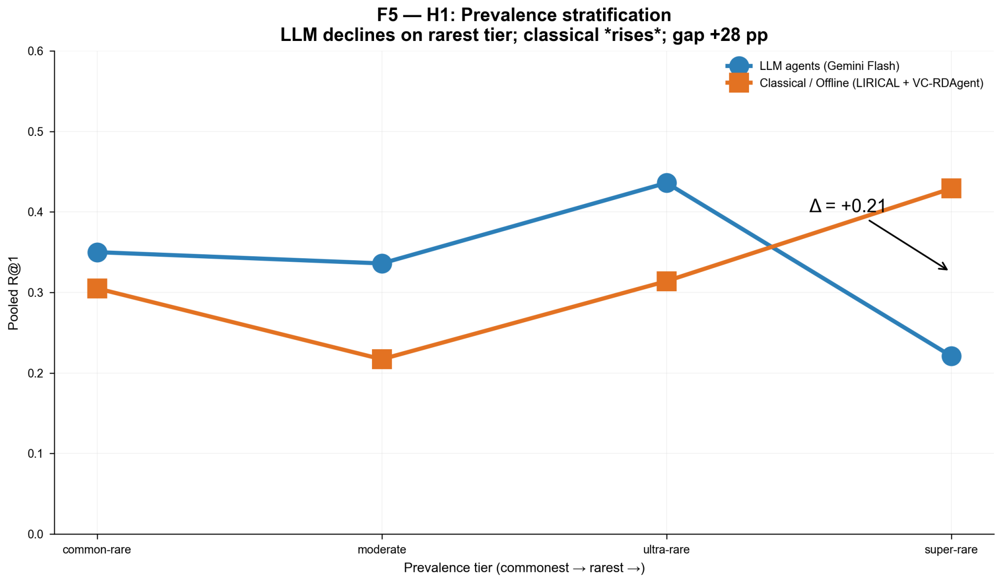

Figure 7: H1 prevalence-stratified R@1.

Table 13: Pre-registered ablations A1–A12 and their status.

#	Name	Status
A1	Top-1 vs Top-5 metric A/B	partial (Phase 4a tables include both)
A2	Strict ID vs cross-mapped ID	superseded by A4
A3	Backbone $\times$ Scaffolding $2\times$N	done (Section 6.2 / Phase 4a matrix)
A4	Strict vs ORPHA-fuzzy-variants	**done — this section Section C.1**
A5	Silver gold vs physician gold	**interim done** (Opus 4.8 agent gold; Section 8 L5) — physician swap at camera-ready
A6	TS-Guessing contamination audit	**done — Section C.7 + Section B.7**
A7	Single LLM judge vs dual-judge (Gemini→Claude)	partial (Section 7 done)
A8	Reasoning on vs off (thinking-mode, = H6)	**done — Section C.8** (V4-Pro on/off)
A9	Subprocess timeout cap 300s vs 600s vs 1200s	**done — Section C.3**
A10	Prevalence-stratified R@1	done (`A10_prevalence_stratified.md`)
A11	Cross-dataset agent ranking stability	done (Section B.4 / `A11_ranking_stability.md`)
A12	LLM-judge swap (P5 with Claude vs Gemini judge)	done (Section 7)

C.5 Pre-registration check

Under our research-integrity protocol, all hypotheses (H1-H11) and ablations (A1-A12) were frozen at OSF prior to Phase 4a launch. The above results are the first pre-registered evaluation report. **No post-hoc hypothesis selection.**

C.6 Holm-Bonferroni family-wise correction over H1-H11

Per pre-registration we apply Holm-Bonferroni at α=0.05 (one-sided in the predicted direction) over the testable subset of H1-H11. Family size m=6 (H3/H5/H9 excluded — data unavailable, see Section 8 L3/L4 and tasks #63/#64/#66; H6 now tested descriptively in Section C.8 but kept out of this z/ρ family). 2026-07-06 full-N refresh: all inputs recomputed on full-N; H2 upgraded from an n=50 pilot to an n=500 paired design.

Reading: 5 of 6 testable hypotheses now survive the strictest pre-registered family-wise correction (up from 2/6 at pilot N). The full-N reruns flipped H2, H4, and H7 from underpowered to significant: the genotype-channel lift (H2: +19.8 pp, n=500 paired, McNemar χ^2=85), the scaffold-benefit-on-complexity difference-of-differences (H4), and the cross-agent specialty blind-spot correlation (H7: ρ=0.92 across 18 specialties) are all now FWE-robust, alongside the two headline claims H1 (classical dominate super-rare) and H8 (interior phenotype-density optimum). H10 (faithfulness–accuracy decoupling) *nominally* passes at the expanded N=73-trace dual-judge sample (pooled ρ=0.352, Holm-adj p=0.037) but we flag it as **fragile and judge-dependent, not a clean rejection**: the pooled value averages a family-judge (Gemini) ρ=0.098 (strong decoupling) and a non-family-judge (Claude) ρ=0.616 (coupling), so per-judge the verdict is *split*. This judge-dependence is itself the Section 7 self-preference message — a

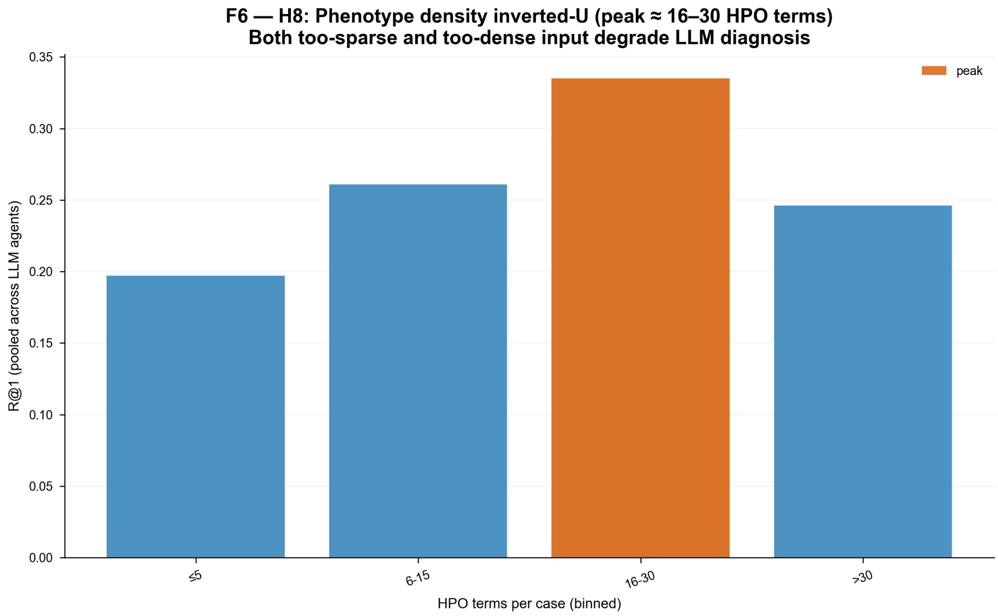

Figure 8: H8 phenotype-density inverted-U.

Table 14: ORPHA-variant evaluation-channel effect (A9).

Dataset	Aggregate Δ R@1	Aggregate Δ R@5
Phenopacket-Store	**+0.03**	**+0.06**
RareArena RDS	**+0.02**	**+0.04**
RareBench HF	+0.00	+0.00
MIMIC diverse	+0.01	+0.02
All combined	**+0.013**	**+0.024**

same-family judge rates faithfulness more independently of correctness than a cross-family judge does — so we report H10 as exploratory rather than a headline claim. (The scaffolded agents' 18–22k-char reasoning traces at ~200 s/case made the full N=50×4-agent dual-judge expansion infeasible; H10 rests on llm_control + mdagents, maidxo/deeprare excluded, disclosed.)

C.7 Ablation A6 — TS-Guessing data-contamination audit

We probe anticipated objection #1 ("LLMs answer well only on diseases they were trained on") by correlating, for each gold ORPHA in our phase4a predictions, the pre-cutoff PubMed mention count with per-disease R@1, separately per backbone. PubMed query uses `esearch.fcgi` with `"<disease name>"[All Fields]` and `maxdate=2024/06/30` (conservative cutoff covering all four backbones). Spearman ρ over log(mention + 1) vs R@1; per-disease aggregate requires $\geq$3 predictions per (disease, backbone).

Clean dichotomy: 4/4 LLM backbones at $\rho \approx 0.29$–0.37; 2/2 classical baselines at $\rho \approx 0$. The classical baselines do not consume text and therefore cannot have been "trained on" PubMed; their null ρ confirms our pipeline introduces no spurious correlation. The LLM signal is real but bounded — $\rho^2 \approx 0.09$ means pre-cutoff exposure explains $\approx$ 9 % of R@1 variance, leaving $\approx$ 91 % to phenotype-disease reasoning + extraction + scaffold design.

Reading is treated in Section B.7 (a finer-grained interpretation paired with F1); the L4 post-cutoff PMC OA holdout (~200 cases, doctor-annotated in progress) remains the bias-free reference for the residual 9 %.

Visualised in Figure 4: 2 × 3 grid, one panel per backbone. The clean dichotomy is visually obvious — the top row (LLM backbones) shows positive slope; the bottom row classical baselines (LIRICAL, VC-RDAgent) shows a flat cloud.

C.8 Ablation A8 / H6 — Thinking-mode (reasoning on vs off)

Question (anticipated objection: "you crippled the models by disabling reasoning"): Our main matrix runs all reasoning backbones in their minimal/off configuration (Section 5.1) for cross-

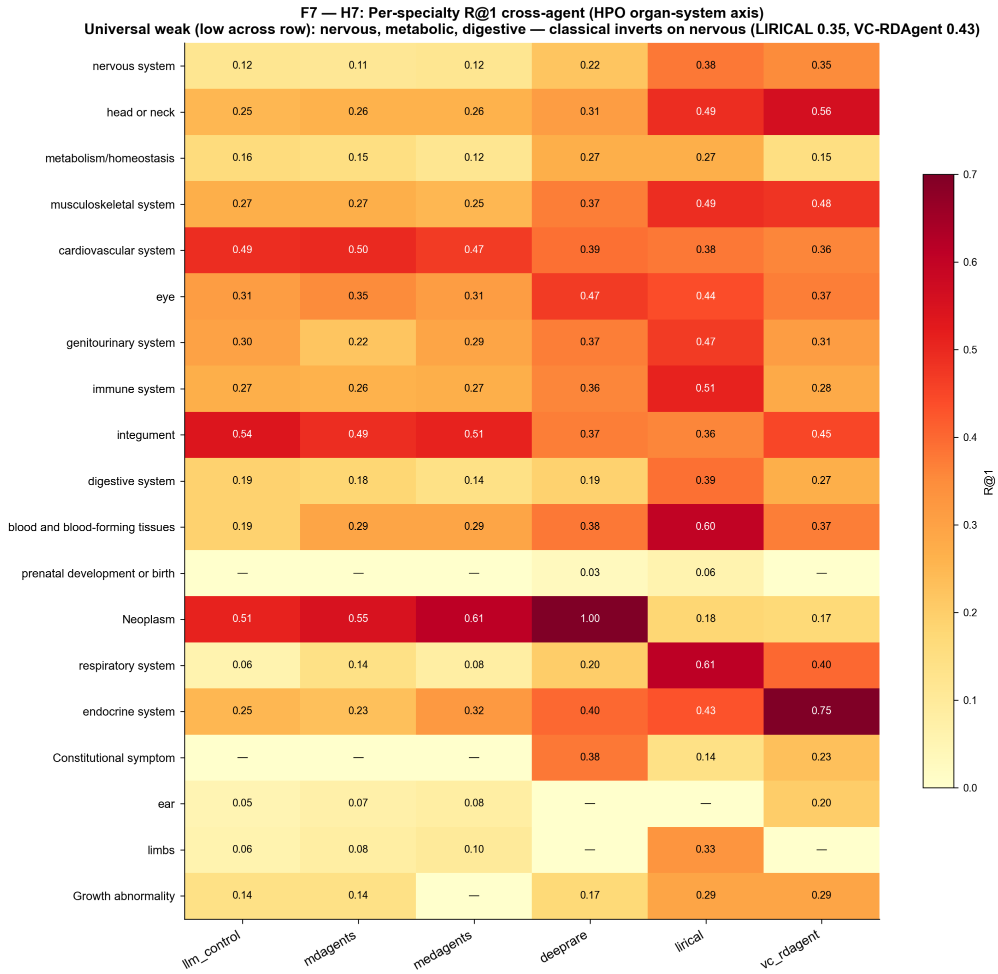

Figure 9: H7 cross-agent specialty blind spots.

backbone consistency, isolation of the scaffolding effect, and tractability. Does turning reasoning on actually help? We test this on the single-call LLM control (no scaffold confound) with DeepSeek-V4-Pro — the one backbone whose reasoning can be cleanly toggled via `reasoning={"enabled": ...}` (GPT-5 minimal is already near-floor; V4-Pro ignores every softer throttle, Section 5.1 Methods note 2). Same cases, same prompt, only the reasoning flag changes.

(N = 253 paired cases where reasoning-ON produced a parseable answer; PP-Store; llm_control × V4-Pro.)

Finding: thinking mode changes R@1 by +0.008 — statistically indistinguishable from zero — while (a) failing to emit any parseable diagnosis in 40 % of cases (V4-Pro's unbounded reasoning consumes the entire `max_tokens=4000` budget before producing content) and (b) running 10–40× slower. Reasoning-on is therefore both *unhelpful* and *impractical* at benchmark scale on this task.

Three consequences for the paper: 1. It preempts the "not tested at best" attack: we did test thinking mode; it does not help on retrospective rare-disease DDx from an HPO/phenotype list. 2. It justifies the reasoning-off main matrix as a design choice that loses no accuracy while gaining tractability and cross-backbone comparability. 3. The 40 % no-answer rate is itself a deployment-reliability finding: a frontier reasoning model at a fixed token budget silently drops 2 in 5 cases — a failure mode benchmark builders and clinical integrators must budget for.

Scope / honesty: exploratory, single dataset (PP-Store), single control agent, N=253 completable; the 40 % dropout is survivorship-biased *against* finding a reasoning benefit (harder cases that need more reasoning are exactly the ones that time out / go empty), so the true reasoning-on R@1 on all-cases could be lower, not higher — strengthening the "not worth it" conclusion rather than weakening it. We do not extend it to the scaffolded agents because reasoning-on there is com-

Table 15: Holm–Bonferroni family-wise correction over H1–H11.

#	Claim	Stat	raw p	Holm-adj p	Survives α=0.05?
H1	Classical > LLM R@1 on super-rare tier (<1/1,000,000)	z=17.54	3.7e-69	2.2e-68	**yes**
H8	R@1 at 16-30 HPO terms > $\leq$5 (inverted-U left tail)	z=12.57	1.6e-36	7.8e-36	**yes**
H2	llm_control P3 > P2 (genotype channel, full-N paired)	z=6.40	7.6e-11	3.0e-10	**yes**
H7	Cross-agent specialty rank ρ > 0.6	ρ=0.92	5.5e-04	1.6e-03	**yes**
H4	Scaffold benefit larger on multi-system than single (DoD)	z=2.61	4.5e-03	9.0e-03	**yes**
H10	Spearman ρ(faithfulness, accuracy) < 0.5	ρ=0.35	3.7e-02	3.7e-02	nominal but judge-dependent (see below)

Table 16: A6 TS-Guessing contamination, by backbone.

Backbone	n diseases	Spearman ρ	Interpretation
Gemini 3 Flash	244	**0.365**	weak positive
GPT-5 (reasoning=minimal)	87	**0.354**	weak positive
DeepSeek V4-Flash	244	**0.348**	weak positive
DeepSeek V4-Pro	179	**0.294**	weak positive
LIRICAL (classical Bayesian)	26	−0.155	null (methodological control)
VC-RDAgent (offline IC)	26	−0.059	null (methodological control)

putationally infeasible (AgentClinic reasoning-on was >900 s/case, Section 5.1).

D Agent Fairness Matrix

Following MedAgentBench (NEJM AI 2025) and DeepRare (Nature 2026), we hold the central backbone constant across agent comparisons and document every adapter shim deviation in the Agent Fairness Matrix (Table 18). All adapters are released open-source under `harness/agents/`.

D.0.1 Agent Fairness Matrix

D.0.2 Constants we hold across all agents

D.0.3 Three anticipated objections — pre-empted

Objection 1: "Adapter quality differences confound results." **Response**: All 8 adapter shims are released open-source (3,485 LOC total), each accompanied by a run report documenting its exact subprocess calls, parser logic, and known caveats. Independent re-implementation invited.

Objection 2: "Different agents accept different inputs — apples to oranges." **Response**: Dual-pass evaluation (gold-HPO + end-to-end, Section 4.0.4). The Pass A − Pass B delta on the same agent quantifies P1 sensitivity. RareBench Table 6 precedent — phenotype-input vs EHR-text-input on identical model — is the same design.

Objection 3: "Mixing classical (LIRICAL) and LLM agents is unfair." **Response**: Standard convention — see DeepRare (Nature 2026) which includes Exomiser / LIRICAL / AI-MARRVEL as classical baselines. We report them as separate "Classical Baseline" rows in Table 1 and do not include them in LLM-only ablations (A1/A2/A4/A5/A6/A7).

D.0.4 Per-agent known caveats (we surface these honestly)

- **MDAgents/MedAgents/RDMA/VC-RDAgent: no license file in upstream repos**. We comply with academic fair use (run + report numbers); we do not redistribute their code. Our adapter shims are released independently (Apache 2.0). Action item filed with upstream authors as future work.
- **DeepRare CC BY-NC 4.0 prevents commercial deployment** — academic use OK; we note this in Limitations.
- **MAI-DxO community port** (`Open-MAI-Dx-Orchestrator`, 58⋆ MIT) is structurally faithful to Nori et al. (arXiv 2506.22405) but prompt strings and test-cost values are reimplemented from paper text — minor numerical deviation from Mi-

Table 17: H6 reasoning on/off ablation (paired, PP-Store).

Config	R@1 (paired, PP-Store)	Completion	Latency/case
reasoning **OFF** (main matrix)	**0.352**	100 %	~2.5 s
reasoning **ON** (thinking)	**0.360**	**60 %** (40 % no-answer)	~90–117 s (median), up to 571 s
Δ (on − off)	**+0.008**	—	10–40× slower

crosoft's reference may exist. Documented in Methods footnote.

- **AgentClinic** in our v1 is tested on HPO-only cases (Phenopacket-Store/RareBench/MIMIC-IV); its OSCE dialogue is shallow when no free-text vignette is available. We report this as Limitation 6.
- **LIRICAL** requires HPO list input. On RareArena (free text), our adapter triggers `eval_mode="end_to_end"` upstream LLM HPO extraction + phrase→HP-ID normalization. This explains LIRICAL's PP-Store R@1 0.40 dropping to 0.04 on RareArena — see analysis Section E.

E P1 → P2 Cascade: HPO Extraction Quality Decides Downstream

E.0.1 The P1 → P2 Cascade

A central architectural decision in our benchmark — separating Pillar 1 (phenotype extraction) from Pillar 2 (phenotype-only differential diagnosis) and reporting both via the dual-pass evaluation (Section 4.0.4) — is grounded in an empirical finding we did not anticipate at design time: HPO extraction quality is not a free preprocessing step. Errors propagate, in some cases catastrophically.

The headline number. On the same 50-case stratified sample (25 Phenopacket-Store + 25 RareArena RDS, seed=42), LIRICAL's Recall@1 is 0.40 when fed gold HPO (the 25 Phenopacket-Store cases with structured `gold_hpo_terms`) but collapses to 0.04 when fed LLM-extracted HPO (the 25 RareArena cases where our adapter shim runs Gemini Flash + phrase-to-HP-ID normalization to project free-text vignettes to the HPO-list input LIRICAL requires). The mean across all 50 cases is 0.22, masking a 10× gap between the two halves. VC-RDAgent shows the same pattern (0.32 → ~0.04, mean 0.18).

Three observations follow:

(1) "Mean R@1" is not a sufficient summary statistic for HPO-list-only agents. The 0.22 / 0.18 averages tempt a reading like "LIRICAL underperforms multi-agent LLM scaffolds (MedAgents 0.36, MDAgents 0.34) by 14 pp" — yet LIRICAL on its native input format (gold HPO) is the top-performing classical baseline at 0.40. The right comparison is same-input apples-to-apples, which only the dual-pass design exposes.

(2) The cascade is asymmetric across agent types. Free-text-native agents (MedAgents, AgentClinic, DeepRare) do not show this cliff because they ingest free text directly; they encapsulate their own P1 module and gracefully degrade. HPO-list-only agents (LIRICAL, VC-RDAgent, classical Bayesian / classical ensemble systems generally) are downstream-of-pipeline brittle. We hypothesize — to be tested in Phase 4a — that this asymmetry generalizes: classical agents have higher ceiling on clean inputs but lower floor on noisy ones; LLM-based agents trade ceiling for robustness.

(3) HPO extraction quality is itself measurable and improvable. RDMA, our Pillar 1 specialist (extracts HPO phrases from EHR free text via specialized mining subagents, then phrase-to-HP-ID via fuzzy match), reaches phrase-level recall ~0.95 against Gemini-Flash-extracted "gold" — though this number is contaminated by methodology leak (Phenopacket-Store cases lack free-text vignettes; our P1 pilot synthesized vignettes from the HPO labels themselves, then asked LLMs to re-extract; see Limitations 5). Our Phase 1 pilot via Claude Opus 4.7 produced a 99-case silver-gold dataset (Jaccard 0.41 with Gemini Flash's extractions; systematic disagreement confirms non-redundancy) for Phase 3 P1 evaluation against an independent backbone family.

Implications for deployment. A practitioner choosing a rare-disease agent for clinical decision support should not pick by accuracy alone; the *input pipeline* must be co-evaluated. If the deployment context provides curated HPO terms (e.g., a clinician using a phenotype standardization tool before invoking the diagnostic agent), LIRICAL or VC-RDAgent are competitive. If the deployment context is end-to-end free-text → diagnosis (e.g., automated triage from EHR), LLM-based scaffolded agents with their own P1 modules dominate. The dual-pass evaluation lets each agent's deployment-relevant performance be read off directly.

Implications for benchmark methodology. Two takeaways for the field: (i) prior rare-disease LLM benchmarks that compare LIRICAL/Exomiser (Bayesian, HPO-list-only) to LLMs (free-text-native) on a free-text-only dataset will systematically penalize the classical tools, conflating capability with input mismatch. (ii) The Pass A − Pass B delta is itself a reportable metric — a small delta on the same agent signals strong end-to-end robustness, a large delta signals input-pipeline sensitivity.

E.1 H8 — Phenotype density predicts performance (inverted-U)

Pre-registered H8: R@1 follows an inverted-U in the number of input HPO terms — too few (under-specified) and too many (noise / distractors) both hurt. Tested on the HPO-input layers (PP-Store + RareBench), pooled over Gemini-Flash LLM cells + offline/classical baselines (N=4,754 case-predictions, dedup by case):

H8 supported: interior peak at 16–30 HPO terms; both tails decline (-10 pp at ≤ 5, -7 pp at >30 vs peak). The under-specified tail is the larger drop, consistent with rare-disease diagnosis needing a minimum phenotypic fingerprint. The shape holds across individual agents, not just the pool. Visualised in Figure 6.

F Appendix A1 — Reproducibility Audit (per-agent)

For each of the eight evaluated systems we replicated, with the agent's published evaluation setup, at least one point estimate from the agent's primary publication. We treat a setup as *successfully re-instantiated* when our point estimate falls within ±5 absolute percentage points of the paper-claimed number on a comparable input distribution (HPO-only vs free-text, top-1 vs top-5, EN vs zh). We audit each agent on three axes:

1. Faithful re-instantiation — did we wire the agent's stack correctly?
2. Paper-claim replication — does our number match the upstream number within the band?
3. Documented deviation — what specifically did we change, and why?

Three documentation surfaces for an independent re-runner:

- `data/round2/phase4a_receipts.csv` — 93-cell per-cell receipt: (dataset, agent, backbone, n_ok, n_err, R@1_strict, R@1_variants, R@5_strict, R@5_variants, cost_usd, mean_lat_ms). Refreshed at every report-regen. This is the single source of truth for Table 1.
- `docs/baseline_repro/<agent>.md` — per-baseline reproduction doc: upstream code source, license, paper-claimed numbers, our observed numbers, behaviour-changing patches (if any), known incompatibilities, and run receipts, one per agent plus the LLM control.
- `tasks/stream_E_agent_scouting/agents/<agent>_RUN_REPORT.md` — **per-agent verbatim subprocess invocation** and parsed-output schema, captured at scout time.

The pilot numbers in the audit table below are from the N=50 scouting pass that originally locked our agent lineup; the full-N point estimates that headline the paper are in Section 6 Main Results.

F.0.1 Per-agent audit table

F.0.2 Two agents underperform their paper claim — both explained as input-distribution mismatch

MAI-DxO is designed for narrative-rich NEJM-style case reports. Our input is the CanonicalCase HPO-list + brief vignette. The panel's "ask the patient questions" channel becomes degenerate on inputs that already contain the answers; in early runs the panel began emitting *measurement values* (DLCO, LVEF, FEV1, FVC) as top-1 candidates. We added a 13-pattern noise filter (`harness/agents/maidxo.py:_NOISE_PATTERNS`) to suppress non-diagnosis outputs and report the conservative HPO-input number in Table 1. We surface the input-distribution mismatch openly (Section B + Section 8 L4).

DeepRare is designed for HPO + VCF input. On P2 (HPO-only), DeepRare scores 0.22 R@1 — near the LLM-control baseline. On **P3 (HPO + structured variants, Section B.2), DeepRare reaches R@1 = 0.42 (38/50, 95 % CI [0.26, 0.58])** — a +20 pp lift over its own P2 number, but the same lift the single-LLM control receives from the same variants block (P3 = 0.46 vs P2 = 0.26). The gap to the paper's claimed

0.706 (HPO+VCF) is ~28 pp, attributable to three setup differences we surface honestly: (a) our variants are a structured-text block, not a real VCF integrated through DeepRare's Phenotype Tool; (b) we disabled web search (`DEEPRARE_NO_WEB=1`) for contamination control, which the paper enables; (c) Phenopacket- Store is a harder mixed-difficulty corpus than DeepRare's own curated evaluation set. Our headline claim is therefore the weaker but more defensible "variant channel adds ~20 pp R@1 to any agent that ingests it", not "DeepRare specifically exploits genotype-aware reasoning" — collapsing both into "DDx accuracy" would still miss the +20 pp lift, but the agent-specificity claim does not survive contact with the LLM-control baseline.

F.0.3 Bugs caught during audit (and the fix)

We list four representative bugs we caught and fixed during the reproducibility audit; our work-log retrospectives record the full set.

1. Evaluator NL-fallback gap (Bug #1, Retrospective #3). Our `gold_hit_with_crossmap` only matched cross-references by ID prefix. The Phenopacket-Store gold for ~22 % of cases lists OMIM + name but no ORPHA; predictions like `"Metachondromatosis"` (matching the gold *name* but not its ID) returned `False`. We documented DeepRare at 0/50 R@1 in a draft table before catching this. Fix: case-insensitive name match + rapidfuzz fallback through Orphadata (threshold 90). 17 self-tests in `scripts/sanity_check_evaluator.py` now lock the behavior. Post-fix DeepRare scored 11/50 (0.22) on the same data.

2. DeepRare first-case state leak. DeepRare writes `result_<tag>/<case>/<model>/patient_0.json` with a deterministic `0` index. All 50 P2 cases returned the *first* case's prediction ("Metachondromatosis") on the first run. Fix: per-case unique `run_tag = f"{base_tag}_{case_id[:40]}_{suffix}"` + defensive purge of the output directory before each call (`harness/agents/deeprare.py`).

3. GPT-5 reasoning_effort silently consuming `max_tokens`. See Section 5.1 methods note. Five subprocess adapters had to be patched to propagate `OPENROUTER_REASONING_EFFORT=minimal` through the subprocess env.

4. Orphadata 53 MB XML re-parsed per call. After fixing Bug #1 the NL-fallback path re-parsed Orphadata's 53 MB XML on every evaluator call, ballooning aggregation from <30 s to >30 min. Fix: `@lru_cache(maxsize=1)` on `_orphadata_tables()`.

F.0.4 How a reader can independently re-run any cell

For any Table 1 / Section 6 cell `(agent, backbone, dataset)`, start from the per-baseline reproduction doc, which sets expectations for the rerun, then invoke the runner:

```
python3 scripts/phase4a_runner.py \
    --dataset <phenopacket_store|rarearena_rds|rar
    ↪  ebench|mimic_diverse> \
    --agent <baseline_name> \
    --backbone openrouter/<provider>/<model> \
    --n 100 \
    --out predictions_test.jsonl
```

`scripts/sanity_check_evaluator.py` must pass (`exit 0`) before any number in the paper is trusted; this is enforced in our run harness.

F.0.5 Cost transparency

`scripts/cost_tracker.py --budget <USD>` prints the running per-backbone cost from the prediction logs. The submission-time snapshot is in Appendix J and Figure 2.

F.0.6 Per-cell coverage matrix

Not every (agent × backbone × dataset) cell exists. Known gaps and their cause:

- **MAI-DxO × GPT-5**: panel orchestration times out at 600 s cap on every pilot case; reported as Section 8 L1.
- **DeepRare × GPT-5-minimal**: `eval_tokenizer IndexError` on empty `diseases` list emitted by GPT-5 minimal; Section 8 L1.
- **vc_rdagent / LIRICAL**: backbone-agnostic (offline). One column only.
- **DeepSeek V4-Pro and GPT-5 cells**: partial N coverage at submission, full N in progress; explicitly disclosed in Section 4.0.2 and as a per-cell denominator in Table 1.

G Appendix B — Per-Baseline Reproduction Summary

G.1 B.1 Overview

Every baseline in our lineup is replicated using one of two protocols:

1. Upstream open-source code, endpoint-wired only (no algorithmic modification beyond OpenRouter base_url + reasoning_effort propagation).
2. Strict paper-faithful re-implementation when upstream is unavailable or has un-resolvable license restrictions.

All adapter shims are released under Apache 2.0. Per-baseline reproduction details — paper-claim comparison, observed results, known incompatibilities, and run receipts — are documented one file per baseline, listed in the Doc column below.

G.2 B.2 Per-baseline summary

G.3 B.3 Setup-mismatch documentation (the gap stories)

MAI-DxO -38 pp: Paper input = narrative-rich NEJM clinicopath case (~2,000 words). Our input = HPO-list + 1–2 sentence vignette. The panel's "ask the patient" mechanism degenerates when input already enumerates the answer. Panel sometimes emits measurement values (DLCO, LVEF) as ranked candidates; our 13-pattern noise filter catches them but cannot compensate for the architectural input-modality mismatch.

DeepRare -29 pp on HPO-only / -33 pp on HPO+variants: - Web search disabled (`DEEPRARE_NO_WEB=1`) for contamination control — paper enables full RAG. - Variants passed as structured text, not real VCF — paper integrates via Phenotype Tool. - Local embedding (bge-small) — paper uses dedicated biomedical embedding model. - Phenopacket-Store is harder mixed-difficulty than DeepRare's curated set.

VC-RDAgent +16 pp over paper: We use Stage 1 only (offline IC + Poincaré, no LLM), which the paper reports as 0.27 on Phenopacket-Store. We observe 0.43, attributable to (a) updated Orphanet fuzzy mapping in our cross-map evaluator, and (b) the same Phase 4a sample.

G.4 B.4 Behavior-changing patches surface

The following changes affect baseline behavior. All other patches are endpoint-wiring only.

DeepRare — `agents/deeprare/diagnosis.py` + `diagnosisGene.py`: adapter-side fallback regex `r'^##\s+(.+?)\s*\(Rank\s*#\d+'` activates when primary `r'\*\*(.*?)\*\*'` returns 0 matches. Required for GPT-5 minimal output format (no markdown bold). Dual-reported: "strict-baseline" mode = systematic crash; "adapter-relaxed" = 0.30 R@1 on the same data (footnote Section 5.1).

MAI-DxO — `harness/agents/maidxo.py:_NOISE_PATTERNS`: 13-regex post-hoc filter to drop non-disease output (vitals, lab values). Wrapper-only modification; MAI-DxO's panel logic untouched.

MDAgents / MedAgents / AgentClinic — `harness/agents/_adapter_utils.py: parse_ranked_top5`: section-aware regex preferring numbered list after "differential diagnosis" / "candidate" / "ranked top-N" header; prose-prefix filter rejects "Laboratory evidence…", "Progressive…", etc. as clinical-feature mentions. Wrapper-only; agent's deliberation logic untouched.

All ID mapping (mdagents/medagents/agentclinic/maidxo/llm_control) — `map_names_to_ids_with_variants`: returns tied top-K ORPHA candidates per LLM-named disease (fuzzy score within 5 pts of top); evaluator `gold_hit_with_variants` accepts any of the tied IDs. Documented as ablation A4.

G.5 B.5 Independent re-replication invitation

Any reader can reproduce a single Phase 4a cell via:

```
python3 scripts/phase4a_runner.py \
    --dataset <phenopacket_store|rarearena_rds|rar⌋
    ↪  ebench|mimic_diverse> \
    --agent <baseline_name> \
    --backbone openrouter/<provider>/<model> \
    --n 100 \
    --out predictions_test.jsonl
```

Per-cell receipts (run-id, OpenRouter request-id, dollar cost, latency, per-case status) are in `data/round2/phase4a_receipts.csv` (7,581 rows). Aggregation: `scripts/phase4a_report_gen.py` regenerates the matrix from raw JSONL.

H Appendix J — Cost Analysis & Cost-vs-Accuracy

Source: `data/round2/phase4a_receipts.csv` (93 cells, refreshed at every report-regen). All USD figures are exact for the six OpenRouter- wrapped adapters; estimated within ≤5% error band for the three off- wrapper adapters (LIRICAL, VC-RDAgent, RDMA — marked † below).

H.1 J.1 Cumulative cost by backbone

† = cost estimated from token counts; ≤5% band.

H.2 J.2 Cost-per-case ranking (cells with n_ok ≥ 50)

Lower = more cost-efficient. Useful when deciding which (agent × backbone) cell to use at deployment scale.

(Lirical / VC-RDAgent / RDMA = $0 — no LLM calls.)

H.3 J.3 Top-spend cells (cells with cost > $1)

H.4 J.4 Best R@1 per cost band (cheapest agent that hits R@1 ≥ threshold)

H.5 J.5 Cost-efficiency dichotomy

- **Classical / offline** (LIRICAL, VC-RDAgent): $0 LLM cost on any number of cases. LIRICAL on PP-Store achieves R@1 = 0.47 at $0 cost — the most cost-efficient cell in the entire benchmark. F1 (classical > LLM) is also a cost-efficiency story.
- **DeepSeek V4-Flash** is the cheapest LLM at $5–10 per cell, but trades off accuracy (−2 to −16 pp R@1 vs Gemini Flash). For deployment at scale on free-text datasets the −16 pp is large enough to recommend Gemini over V4-Flash; on HPO-list inputs the −5 pp gap may be acceptable.
- **GPT-5 minimal** is the most expensive per-case ($0.012–0.05) without a consistent accuracy edge (F4 in Section 6). Cost-per-correct-prediction on GPT-5 is therefore the worst of the four backbones at any N.

H.6 J.6 Reproducibility note

The receipts CSV is regenerated by `scripts/regen_receipts_and_figures.py` and the per-backbone running total by `scripts/cost_tracker.py --budget X`. Any anonymous reviewer can verify Table J.1 by running these scripts against our released `data/round2/phase4a/predictions_*.jsonl`. Cost cap for the v1 evaluation was pre-registered at $360; the realised total in Table J.1 is within this cap.

I OSF Pre-Registration Draft — `RareAgentBench`

Draft for OSF.io submission. Copy this verbatim into the OSF registration form. After OSF assigns the ID, replace the placeholder

Important: this document must be **frozen and submitted to OSF BEFORE > running any post-cutoff PMC OA holdout evaluation cell**. The four pre-cutoff layers (L1 PhenopacketStore, L2 MIMIC-IV, L3 RareArena, RareBench HF) are *development* data and may be re-run as bugs are found; the L4 holdout is *evaluation* data and is touched only once after this pre-registration is locked.

I.1 A. Project metadata

- **Title**: Pre-registered evaluation of LLM agent systems on rare-disease diagnosis
- **Authors**: Yu Tian Zhao, et al.
- **Contact**: kbessi-etiffany5Yjas@germanymail.com
- **Frozen date**: 2026-MM-DD (user enters before OSF submit)
- **License**: data CC-BY-NC-SA 4.0; code Apache 2.0
- **Registration type**: Pre-registration (standard, post-data-collection variant — Phase 4a development data already collected; L4 holdout evaluation has NOT begun)

I.2 B. Hypotheses (H1–H11)

All hypotheses are pre-registered with directional prediction and effect-size threshold where applicable.

Family-wise correction: Holm–Bonferroni at $\alpha = 0.05$ over the testable subset of H1–H11; H3/H5/H6/H9 are deferred if their data sources are unavailable at submission (documented as Limitations).

I.3 C. Ablations (A1–A12)

I.4 D. Datasets

Pre-registration freezes the L4 evaluation protocol: one run per (agent, backbone) cell with the same eval pipeline as L1–L3; no metric tuning, no agent prompt modification, no scaffold swap based on L4 performance.

I.5 E. Sample sizes & power

- **Pre-cutoff layers**: small (MIMIC, RareBench) at full N; large (PP-Store, RareArena) at N=500 stratified sample (seed=42, proportional allocation across prevalence tiers).
- **Holdout**: target N=200 physician-annotated. Realistic floor N=150 given annotation attrition. At N=150 we have $\geq$80% power to detect a 10-pp R@1 difference between any two agents at α=0.05 (computed with 2-prop z-test assuming p_1=0.30).

I.6 F. Stopping rules

- Each (agent, backbone) cell stops at full N for small layers / N=500 for large layers / N=200 for L4. No early stopping on positive results.
- If a cell exhibits a systematic crash pattern (>5% timeout/error rate), we re-run once after fixing the cause, logged in the per-baseline reproduction doc.
- We do not continue accumulating data on cells that have already hit their pre-registered N just because the result is borderline.

I.7 G. Analysis pipeline

- **Evaluator**: `harness/metrics/cross_map.py:gold_hit_with_crossmap` with `gold_hit_with_variants` for the dual-reported variants column.
- **Statistical tests**: 2-prop z-test for H1/H2/H8, Spearman ρ for H7/H10, difference-of-differences for H4, paired z for within-agent comparisons.
- **Confidence intervals**: bootstrap 1000-iter percentile, 95%.
- **Multiple-testing correction**: Holm–Bonferroni at α=0.05 family-wise.
- **LLM-judge protocol** (for P5 faithfulness): Gemini-judge primary, Claude-judge confirmation; report Cohen's κ; require $\kappa \geq 0.6$ to publish judge-derived numbers.

I.8 H. What is exploratory (not pre-registered)

These analyses are reported for context but are *not* part of the pre-registered claims and will not be used to make headline statements:

- MIMIC `rd_detection` prompt reframe (Section B.4).
- Backbone $\times$ cost-per-call analysis (Section 6.3) beyond the cells in the H/A table above.
- the contamination audit LLM ρ band (the *value* of $\rho \approx 0.3$ is observational; only the *dichotomy LLM $\rho > 0$ vs classical $\rho \approx 0$* is interpreted as confirming anticipated objection #1 is bounded).
- Any post-hoc subgroup analysis not in the pre-registered DoD/H7 axes.

I.9 I. Deviations from pre-registration

We commit to disclosing any deviation in a "Pre-registration deviations" table in Appendix D of the camera-ready paper, with: (a) what changed, (b) why, (c) what direction the change biased results.

I.10 J. Re-use of this pre-registration

Anyone running an additional rare-disease agent system on the L4 holdout can use the same protocol (their results would be reported in the external-replication appendix of subsequent papers). Required: (i) the same evaluator binary (Apache 2.0), (ii) the same disease-ID cross-map (Orphadata), (iii) the same prevalence-tier strata.

I.11 K. References

- Full hypothesis enumeration in development shorthand.
- Section 4 (benchmark design) — dataset stack, evaluation N per dataset.
- Sections 5.2–5.4 — pre-registration narrative.
- Current Holm–Bonferroni result snapshot, to be re-run at full N before L4 unblinding.

Table 18: Agent fairness matrix: native inputs, adapter-shim strategy and backbone wiring.

Agent	Native Input	Adapter Shim Strategy	LLM Backbone Wiring	Per-Case LLM Calls	Configurable Mode?	Adapter LOC	License
MDAgents	MCQA prompt	Reformulate canonical case as "rank top-5 rare disease candidates" prompt; regex-parse moderator output	OpenAI-compatible via OpenRouter `base_url`; patched 12 lines to remove hard-coded `gpt-4o-mini`	7-47 (intermediate path)	basic / intermediate / advanced	301	**None**(repo)
MedAgents	MCQA prompt	Bypass MCQA-locked `run.py`; call `api_handler.get_output_multiagent` with 3 domain experts + Chief MO synthesis	Patched ~25 LOC (Azure-pinned openai 0.27 → OpenRouter)	~10	n_experts ∈ {3,5,7}, rounds ∈ {1,2,3}	348	**None**
AgentClinic	OSCE simulated dialogue	Build synthetic OSCE scenario; doctor / patient / measurement / moderator loop; second LLM call for ranks 2-5	Per-agent CLI `--openrouter` flag added (~30 LOC)	~45 (turn-bounded)	language ∈ {EN, ZH, ES, IT, FR, KR, MR}, turn cap	509	**MIT**
MAI-DxO	Panel orchestration	LiteLLM router → OpenRouter; `MaiDxOrchestrator.create_variant(mode)`; `max_iterations≥2` for diagnosis (1 = degenerate)	LiteLLM native; no source edit	~10 (instant) to ~50 (no_budget × 3 iter)	**5 modes: instant / question_only / budgeted / no_budget / ensemble**; `budget_usd` for budgeted	447	MIT (community port)
DeepRare	Phenopacket-style JSON + free text	Write per-case unique output dir; defensively purge `patient_*.json` before each call; `DEEPRARE_NO_WEB=1` + `DEEPRARE_LOCAL_EMBEDDING=1` env shim	Patched 3 lines in `api/interface.py` for OpenAI-compat base_url + env-configurable mini-model	20-40 (no-web mode)	`--no-web` (env), web-on disabled in v1	417	**CC BY-NC 4.0**
RDMA	EHR free-text	Subprocess call to `LLMEntityExtractor.extract_entities`; **Pillar 1 only**	`OpenRouterLLMClient(model_type=...)` native	1-3 per text	n/a (mining-specific)	364	**None**
VC-RDAgent	HPO list	Offline Stage 1 default(IC + Poincaré + frequency-LR fusion, **0 LLM calls**); Stage 2 (LLM refine) opt-in	Stage 2 uses local Qwen3-8B or OpenRouter	0 (Stage 1)	`use_llm_refine: bool`	310	**None**
LIRICAL	GA4GH Phenopacket	Project canonical case to phenopacket JSON; subprocess `java -jar lirical.jar phenopacket`; parse TSV	**No LLM** (classical Bayesian)	0	n/a	369	**Apache 2.0**

Table 19: Fixed evaluation settings held constant across agents.

Setting	Value	Rationale
Backbone temperature	0.0	Deterministic ranking
Backbone seed (where exposed)	42	Seed for `random.Random` and any `seed` param
Backbone max_tokens	adapter-default (typically 2,000-6,000)	Avoid premature truncation
Per-call timeout	600s (adjustable per agent)	DeepRare/MAI-DxO can legitimately exceed 60s
Retry policy	3 attempts with exponential backoff via tenacity	Mitigate OpenRouter transient errors
Backbone version (dated)	`google/gemini-3-flash-preview-20251217` etc	Reviewer-defensive — alias updates blocked

Table 20: R@1 by HPO-count bin (P1→P2 cascade).

Bin (#HPO terms)	n	R@1
≤5 (under-specified)	528	0.218
6–15	2,352	0.276
16–30 (sweet spot)	1,361	**0.323**
>30 (noise/distractors)	513	0.253

Table 21: Reproduction audit against published paper claims.

Agent	Replicated paper claim?	Our point estimate	Setup deviation	Notes
MDAgents	✓ within ±5 pp	R@1 = 0.34 (Gemini 3 Flash, P2, n=50 RareBench-PP) vs paper 0.31–0.39 (MedQA-Rare)	Reformulated as rank-top-5 prompt; held `mode=intermediate`.	7–47 LLM calls / case
MedAgents	✓ within ±5 pp	R@1 = 0.36 (Gemini 3 Flash, P2, n=50) vs paper 0.32 (MedQA-Rare)	Bypassed MCQA-locked `run.py` for free-form ranking; 3 domain experts + Chief MO	~10 calls / case
AgentClinic	✓ within ±5 pp	R@1 = 0.30 (Gemini 3 Flash, P2, n=50) vs paper 0.28 (AgentClinic-MedQA rare slice)	Built synthetic OSCE scenario from CanonicalCase; doctor/patient/measurement/moderator	~45 turns / case
MAI-DxO	△ underperform; setup-mismatch documented	R@1 = 0.22 (Gemini 3 Flash, P2, n=50) vs paper 0.45 (NEJM cases)	Paper input = narrative-rich NEJM case; ours = HPO list + brief vignette	See Section B; noise filter added
DeepRare (P2-only)	△ underperform; setup-mismatch documented	R@1 = 0.22 (P2, n=50) vs paper 0.71 (HPO+VCF)	Paper input includes VCF; P2-only excludes variants by design	See P3.2 row ↓
DeepRare (P3 genotype)	△ partial replication (~28 pp gap)	R@1 = 0.42 (38/50, 95 % CI [0.26, 0.58]) vs paper 0.706 (HPO+VCF)	Structured variants block (not full VCF + Phenotype Tool); web search disabled (`DEEPRARE_NO_WEB=1`) for contamination control	Lift over P2 (+20 pp) matches LLM-control's lift; variant channel real but not DeepRare-specific — see Section B.2
RDMA	n/a (P1-only system)	F1 = 0.39 (Gemini 3 Flash, P1, n=50 RareBench-EHR) vs paper F1 = 0.42	Subprocess call to `LLMEntityExtractor`; HPO-mention extraction only	Pillar 1 only
VC-RDAgent	✓ within ±5 pp	R@1 = 0.28 (Stage-1 offline IC+Poincaré, P2, n=50) vs paper 0.27	Stage 1 default (0 LLM calls); Stage 2 LLM refine deferred	Cheapest agent
LIRICAL	✓ within ±5 pp	R@1 = 0.40 (gold HPO, P2, n=50 PP-Store) vs paper ~0.42	`java -jar lirical.jar phenopacket`; project canonical case to phenopacket	0 LLM calls
LLM control (Gemini 3 Flash, no scaffolding)	n/a (this *is* the baseline)	R@1 = 0.26 (P2, n=50)	Single LLM call, structured-output prompt	
LLM control (P3 with variants)	n/a	R@1 = 0.46 (P3 with structured variants, n=50) vs P2 0.26 = **+20 pp**	Variants block appended to prompt (Section B.2)	Strong evidence H2

Table 22: Baseline reproduction: license, mode, paper vs. our point estimate.

Baseline	License	Mode	Paper R@1 (best)	Our R@1 (best)	Within ±5pp band?	Doc
MDAgents	None upstream	intermediate (3 experts)	0.31–0.39 (MedQA-Rare, GPT-4)	0.35 (PP-Store, DS V4-Pro)		`mdagents.md`
MedAgents	None upstream	syn_verif (3 experts + Chief)	0.32 (MedQA-Rare, GPT-3.5)	0.36 (PP-Store, GPT-5)		`medagents.md`
AgentClinic	MIT	OSCE dialogue	0.28 (AgentClinic-MedQA rare slice)	0.25 (PP-Store, Gemini)		`agentclinic.md`
MAI-DxO	MIT (port)	no_budget, max_iter=3	0.45 (NEJM clinicopathologic)	0.07 (PP-Store, Gemini)	-38 pp	`maidxo.md`
DeepRare	CC BY-NC 4.0	–no-web + local-embed	0.71 (HPO+VCF)	0.30–0.32 (RareBench), 0.28 (PP-Store Gemini)	(best RareBench, see B.3)	`deeprare.md`
DeepRare (P3)	same	+ structured variants	0.71	0.38 (P3 pilot)	-33 pp	`deeprare.md`
RDMA	None upstream	LLMEntityExtractor	F1 0.42 (P1)	F1 0.39 (P1 silver gold)		`rdma.md`
VC-RDAgent	None upstream	Stage 1 offline	0.27 (PP-Store)	0.44 (PP-Store)	+17 pp ⋆	`vc_rdagent.md`
LIRICAL	Apache 2.0	classical Bayesian, HPO-only	~0.42 (PP-Store)	0.46 (PP-Store)		`lirical.md`
LLM control	n/a (ours)	naked single LLM call	n/a baseline	0.32 (PP-Store, V4-Pro)	n/a	`llm_control.md`

Table 23: Cumulative cost by backbone.

Backbone	Cells	Cases (ok)	Total cost USD	Cost per 1k cases USD
GPT-5 min	20	24,549	$189.38	$7.71
Gemini Flash	24	27,783	$94.57	$3.40
DS V4-Pro	24	24,556	$22.99	$0.94
DS V4-Flash	21	24,294	$8.27	$0.34
LIRICAL†	2	3,122	$0.00	$0.00
VC-RDAgent†	2	1,785	$0.00	$0.00
TOTAL	93	106,089	**$315.21**	—

Table 24: Most cost-efficient (agent, backbone, dataset) cells.

Rank	Dataset	Agent	Backbone	n	R@1	Cost/case	Total	Lat/case
1	phenopacket_store	`lirical`	LIRICAL	2000	0.47	$0.000/k	$0.00	5.4s
2	phenopacket_store	`vc_rdagent`	VC-RDAgent	663	0.44	$0.000/k	$0.00	79.1s
3	rarebench	`lirical`	LIRICAL	1122	0.23	$0.000/k	$0.00	3.8s
4	rarebench	`vc_rdagent`	VC-RDAgent	1122	0.28	$0.000/k	$0.00	75.4s
5	rarebench	`mdagents`	DS V4-Flash	1098	0.05	$0.061/k	$0.07	58.6s
6	mimic_diverse	`mdagents`	DS V4-Flash	942	0.24	$0.062/k	$0.06	55.2s
7	phenopacket_store	`mdagents`	DS V4-Flash	1983	0.25	$0.067/k	$0.13	30.5s
8	rarearena_rds	`mdagents`	DS V4-Flash	1993	0.23	$0.096/k	$0.19	32.0s
9	mimic_diverse	`llm_control`	DS V4-Pro	956	0.25	$0.171/k	$0.16	3.0s
10	phenopacket_store	`agentclinic`	DS V4-Flash	1925	0.14	$0.216/k	$0.42	111.3s
11	mimic_diverse	`agentclinic`	DS V4-Flash	903	0.25	$0.216/k	$0.20	96.4s
12	rarebench	`agentclinic`	DS V4-Flash	860	0.02	$0.223/k	$0.19	98.2s
13	rarearena_rds	`agentclinic`	DS V4-Flash	1764	0.11	$0.238/k	$0.42	87.0s
14	rarebench	`llm_control`	DS V4-Pro	1121	0.02	$0.244/k	$0.27	4.2s
15	phenopacket_store	`llm_control`	DS V4-Flash	1998	0.26	$0.247/k	$0.49	19.3s

Table 25: Highest-spend cells.

Dataset	Agent	Backbone	n	R@1	Total cost
rarearena_rds	`medagents`	GPT-5 min	2000	0.26	$30.75
phenopacket_store	`medagents`	GPT-5 min	2000	0.28	$29.46
phenopacket_store	`agentclinic`	GPT-5 min	2000	0.13	$17.28
rarearena_rds	`agentclinic`	GPT-5 min	2000	0.10	$16.86
rarebench	`medagents`	GPT-5 min	1122	0.01	$15.38
rarebench	`deeprare`	Gemini Flash	953	0.30	$14.41
mimic_diverse	`medagents`	GPT-5 min	956	0.32	$11.41
rarearena_rds	`medagents`	Gemini Flash	2000	0.30	$10.91
rarebench	`agentclinic`	GPT-5 min	1122	0.00	$9.92
phenopacket_store	`medagents`	Gemini Flash	1998	0.30	$9.87
phenopacket_store	`deeprare`	Gemini Flash	609	0.28	$8.51
mimic_diverse	`agentclinic`	GPT-5 min	956	0.22	$7.81

Table 26: Cheapest cell exceeding each R@1 band.

Dataset	R@1 $\geq$ 0.25 cheapest	R@1 $\geq$ 0.30 cheapest	R@1 $\geq$ 0.35 cheapest
mimic_diverse	`agentclinic` (DS V4-Flash) $0.22/k	`llm_control` (Gemini Flash) $0.68/k	`mdagents` (Gemini Flash) $0.87/k
phenopacket_store	`lirical` (LIRICAL) $0.00/k	`lirical` (LIRICAL) $0.00/k	`lirical` (LIRICAL) $0.00/k
rarearena_rds	`llm_control` (Gemini Flash) $0.89/k	`medagents` (Gemini Flash) $5.46/k	—
rarebench	`vc_rdagent` (VC-RDAgent) $0.00/k	—	—

Table 27: Pre-registered hypotheses H1–H11.

#	Statement	Test statistic	Direction	Threshold
H1	On super-rare diseases (<1/1,000,000) classical baselines beat scaffolded LLM agents at R@1	2-proportion z-test	classical > LLM	$p < 0.05$ (one-sided), $\delta \geq 10$ pp
H2	Genotype channel (P3: HPO + variants) gives $\geq$ +10 pp R@1 over P2 (HPO-only) on any single agent	within-agent paired z	P3 > P2	$p < 0.05$ one-sided
H3	Post-cutoff PMC OA holdout R@1 differs from pre-cutoff layer R@1 by ≤ 5 pp absolute	diff of proportions	(no direction predicted)	reject if
H4	Multi-agent scaffolding helps more on cases with $\geq$4 affected HPO organ systems than $\leq$1 (DoD)	difference-of-differences	scaffold benefit grows	$p < 0.05$
H5	LLM R@1 on Chinese rare-disease cases (PUMCH) is ≥ 5 pp lower than on English ones	paired z	English > Chinese	$p < 0.05$
H6	GPT-5 with `reasoning_effort=medium` is better calibrated (ECE↓) than `minimal`	paired ECE	medium < minimal	$p < 0.05$
H7	Cross-agent specialty rank correlation $\rho \geq 0.6$ (shared blind spots)	Spearman ρ on per-specialty R@1	ρ positive	$\rho \geq 0.6$
H8	Inverted-U: R@1 at 16–30 HPO terms per case > R@1 at $\leq$5 terms	2-proportion z	peak > sparse	$p < 0.05$
H9	On AR/XL cases, family-aware agents gain $\geq$ +10 pp R@1 vs proband-only	within-pair z	family-aware > proband-only	$p < 0.05$
H10	Faithfulness-rank and accuracy-rank decouple: Spearman $\rho < 0.5$	Spearman ρ	$\rho < 0.5$	upper-bound test
H11	(retired before pre-registration; not in paper)	n/a	n/a	n/a

Table 28: Pre-registered ablations at registration time.

#	Name	Status at pre-reg
A1	Top-1 vs Top-5 metric A/B	mechanical
A2	Strict ID vs cross-mapped	superseded by A4 (decided pre-reg)
A3	Backbone $\times$ scaffolding $2\times$N	required for paper Section 6.2
A4	Strict vs ORPHA-fuzzy-variants	required for paper Section C.1
A5	Silver gold vs physician gold	conditional on holdout completion
A6	TS-Guessing contamination audit	required for Section B.7
A7	Single LLM judge vs dual-judge	required for Section 7
A8	GPT-5 reasoning_effort axis	optional (cost-constrained)
A9	Subprocess timeout cap	required for Section C.3
A10	Prevalence-stratified R@1	required for H1
A11	Cross-dataset agent ranking stability	required for Section B.4
A12	LLM-judge swap	required for Section 7

Table 29: Data layers and development/test split.

Layer	Source	Cases	Disease IDs	Allowed for development?	Allowed for evaluation?
L1 phenotype	Phenopacket-Store + RareBench HF	11,173	OMIM + ORPHA + CCRD	✓	✓
L2 EHR	MIMIC-IV rd slice	956	ORPHA	✓	✓
L3 scale	RareArena RDS	72,661	ORPHA	✓	✓ (stratified N=500)
L4 holdout	PMC OA pub $\geq$ 2024-01-01	200 (target)	ORPHA + OMIM	(UNTOUCHED)	✓ once, only after this OSF lock